

MODULE-6

INNER PRODUCT SPACES

4.1 INTRODUCTION

Inner product space is a vector space with an inner product on it. It associates each pair of vectors in the space with a scalar quantity known as the inner product of vectors. It helps defining the orthogonality between vectors. They generalize Euclidean spaces to the vector spaces of any dimension.

4.2 INNER PRODUCT SPACES

Let V be a real vector space. An inner product on V denoted by $\langle \cdot, \cdot \rangle$ is a function from $V \times V \rightarrow \mathbb{R}$ that assigns a real number $\langle \mathbf{u}, \mathbf{v} \rangle$ to each ordered pair of vectors $\mathbf{u}$ and $\mathbf{v}$ in V in such a way that for all $\mathbf{u}, \mathbf{v}, \mathbf{w}$ in V and all scalars k , the following axioms are satisfied.

- (a) $\langle \mathbf{u}, \mathbf{v} \rangle = \langle \mathbf{v}, \mathbf{u} \rangle$ (Symmetry)
- (b) $\langle \mathbf{u} + \mathbf{v}, \mathbf{w} \rangle = \langle \mathbf{u}, \mathbf{w} \rangle + \langle \mathbf{v}, \mathbf{w} \rangle$ (Additivity)
- (c) $\langle k\mathbf{u}, \mathbf{v} \rangle = k \langle \mathbf{u}, \mathbf{v} \rangle$ (Homogeneity)
- (d) $\langle \mathbf{u}, \mathbf{u} \rangle \geq 0$ and $\langle \mathbf{u}, \mathbf{u} \rangle = 0$ (non-negativity)

if and only if $\mathbf{u} = \mathbf{0}$

If the given product satisfies all the above 4 axioms then V is called a real inner product space with respect to the given product.

Note: If $\mathbf{u} = (u_1, u_2, \dots, u_n)$ and $\mathbf{v} = (v_1, v_2, \dots, v_n)$ are vectors in $\mathbb{R}^n$ then Euclidean inner product (dot product)

4.2.1 *Properties of Inner Products*

If $\mathbf{u}$, $\mathbf{v}$, and $\mathbf{w}$ are vectors in an inner product space V and k is any scalar then

$$(i) \quad \langle \mathbf{0}, \mathbf{v} \rangle = \langle \mathbf{v}, \mathbf{0} \rangle = 0$$

$$(ii) \quad \langle \mathbf{u}, \mathbf{v} + \mathbf{w} \rangle = \langle \mathbf{u}, \mathbf{v} \rangle + \langle \mathbf{u}, \mathbf{w} \rangle$$

$$(iii) \quad \langle \mathbf{u}, k\mathbf{v} \rangle = k\langle \mathbf{u}, \mathbf{v} \rangle$$

$$(iv) \quad \langle \mathbf{u} - \mathbf{v}, \mathbf{w} \rangle = \langle \mathbf{u}, \mathbf{w} \rangle - \langle \mathbf{v}, \mathbf{w} \rangle$$

$$(v) \quad \langle \mathbf{u}, \mathbf{v} - \mathbf{w} \rangle = \langle \mathbf{u}, \mathbf{v} \rangle - \langle \mathbf{u}, \mathbf{w} \rangle$$

Example 1: Let $\mathbf{u} = (3, -1)$, $\mathbf{v} = (2, -2)$, $\mathbf{w} = (-1, 6)$ and $k = -2$. Verify the following using Euclidean inner product.

$$(i) \quad \langle \mathbf{u} + \mathbf{v}, \mathbf{w} \rangle = \langle \mathbf{u}, \mathbf{w} \rangle + \langle \mathbf{v}, \mathbf{w} \rangle$$

$$(ii) \quad \langle \mathbf{u}, \mathbf{v} + \mathbf{w} \rangle = \langle \mathbf{u}, \mathbf{v} \rangle + \langle \mathbf{u}, \mathbf{w} \rangle$$

$$(iii) \quad \langle k\mathbf{u}, \mathbf{v} \rangle = k \langle \mathbf{u}, \mathbf{v} \rangle = \langle \mathbf{u}, k\mathbf{v} \rangle$$

$$(iv) \quad \langle \mathbf{0}, \mathbf{u} \rangle = \langle \mathbf{u}, \mathbf{0} \rangle = 0$$

Solution: (i) $\mathbf{u} + \mathbf{v} = (3, -1) + (2, -2) = (5, -3)$

$$\text{L.H.S.} = \langle \mathbf{u} + \mathbf{v}, \mathbf{w} \rangle$$

$$= (\mathbf{u} + \mathbf{v}) \cdot \mathbf{w}$$

$$= (5, -3) \cdot (-1, 6)$$

$$= -5 - 18$$

$$= -23$$

$$\text{R.H.S.} = \langle \mathbf{u}, \mathbf{w} \rangle + \langle \mathbf{v}, \mathbf{w} \rangle$$

$$= \mathbf{u} \cdot \mathbf{w} + \mathbf{v} \cdot \mathbf{w}$$

$$= (3, -1) \cdot (-1, 6) + (2, -2) \cdot (-1, 6)$$

$$= (-3 - 6) + (-2 - 12)$$

$$= -23$$

$$\text{L.H.S.} = \text{R.H.S.}$$

$$\text{(iii) } k\mathbf{u} = -2(3, -1) = (-6, 2), k\mathbf{v} = -2(2, -2) = (-4, 4)$$

$$\langle k\mathbf{u}, \mathbf{v} \rangle = (k\mathbf{u}) \cdot \mathbf{v}$$

$$= (-6, 2) \cdot (2, -2)$$

$$= (-12 - 4)$$

$$= -16$$

$$k\langle \mathbf{u}, \mathbf{v} \rangle = k(\mathbf{u} \cdot \mathbf{v})$$

$$= -2[(3, -1) \cdot (2, -2)]$$

$$= -2(6 + 2)$$

$$= -16$$

$$\langle \mathbf{u}, k\mathbf{v} \rangle = \mathbf{u} \cdot (k\mathbf{v})$$

$$= (3, -1) \cdot (-4, 4)$$

$$= -12 - 4$$

$$= -16$$

$$\langle k\mathbf{u}, \mathbf{v} \rangle = k\langle \mathbf{u}, \mathbf{v} \rangle = \langle \mathbf{u}, k\mathbf{v} \rangle$$

Example 2: Determine which of the following are inner products on R^3 if, $\mathbf{u} = (u_1, u_2, u_3)$, $\mathbf{v} = (v_1, v_2, v_3)$

(i) $\langle \mathbf{u}, \mathbf{v} \rangle = 2u_1v_1 + u_2v_2 + 4u_3v_3$

(ii) $\langle \mathbf{u}, \mathbf{v} \rangle = u_1^2v_1^2 + u_2^2v_2^2 + u_3^2v_3^2$

(iii) $\langle \mathbf{u}, \mathbf{v} \rangle = u_1v_1 - u_2v_2 + u_3v_3$

Solution: (i) (a) $\langle \mathbf{u}, \mathbf{v} \rangle = 2u_1v_1 + u_2v_2 + 4u_3v_3$

$$= 2v_1u_1 + v_2u_2 + 4v_3u_3$$

$$= \langle \mathbf{v}, \mathbf{u} \rangle$$

Symmetry axiom is satisfied.

(b) Let $\mathbf{w} = (w_1, w_2, w_3)$ be also in R^3 .

$$\mathbf{u} + \mathbf{v} = (u_1, u_2, u_3) + (v_1, v_2, v_3)$$

$$= (u_1 + v_1, u_2 + v_2, u_3 + v_3)$$

$$\langle \mathbf{u} + \mathbf{v}, \mathbf{w} \rangle = 2(u_1 + v_1)w_1 + (u_2 + v_2)w_2 + 4(u_3 + v_3)w_3$$

$$= (2u_1w_1 + u_2w_2 + 4u_3w_3) + (2v_1w_1 + v_2w_2 + 4v_3w_3)$$

$$= \langle \mathbf{u}, \mathbf{w} \rangle + \langle \mathbf{v}, \mathbf{w} \rangle$$

(c) Let k be any scalar.

$$\begin{aligned}k\mathbf{u} &= k(u_1, u_2, u_3) = (ku_1, ku_2, ku_3) \\ \langle k\mathbf{u}, \mathbf{v} \rangle &= 2(ku_1)v_1 + (ku_2)v_2 + 4(ku_3)v_3 \\ &= k(2u_1v_1 + u_2v_2 + 4u_3v_3) \\ &= k\langle \mathbf{u}, \mathbf{v} \rangle\end{aligned}$$

Homogeneity axiom is satisfied.

$$\begin{aligned}\text{(d)} \quad \langle \mathbf{u}, \mathbf{u} \rangle &= 2u_1u_1 + u_2u_2 + 4u_3u_3 \\ &= 2u_1^2 + u_2^2 + 4u_3^2 \geq 0\end{aligned}$$

$$\text{Also, } \langle \mathbf{u}, \mathbf{u} \rangle = 2u_1^2 + u_2^2 + 4u_3^2 = 0$$

if and only if $u_1 = 0, u_2 = 0, u_3 = 0$ i.e. $\mathbf{u} = \mathbf{0}$

Non-negativity axiom is satisfied.

Hence, the given product is an inner product in R^3 .

$$\begin{aligned}
 \text{(ii) (a) } \langle \mathbf{u}, \mathbf{v} \rangle &= u_1^2 v_1^2 + u_2^2 v_2^2 + u_3^2 v_3^2 \\
 &= v_1^2 u_1^2 + v_2^2 u_2^2 + v_3^2 u_3^2 \\
 &= \langle \mathbf{v}, \mathbf{u} \rangle
 \end{aligned}$$

Symmetry axiom is satisfied.

(b) Let $\mathbf{w} = (w_1, w_2, w_3)$ be also in R^3 .

$$\begin{aligned}
 \mathbf{u} + \mathbf{v} &= (u_1, u_2, u_3) + (v_1, v_2, v_3) \\
 &= (u_1 + v_1, u_2 + v_2, u_3 + v_3)
 \end{aligned}$$

$$\begin{aligned}
 \langle \mathbf{u} + \mathbf{v}, \mathbf{w} \rangle &= (u_1 + v_1)^2 w_1^2 + (u_2 + v_2)^2 w_2^2 + (u_3 + v_3)^2 w_3^2 \\
 &= (u_1^2 w_1^2 + u_2^2 w_2^2 + u_3^2 w_3^2) + (v_1^2 w_1^2 + v_2^2 w_2^2 + v_3^2 w_3^2) + 2(u_1 v_1 w_1^2 + u_2 v_2 w_2^2 + u_3 v_3 w_3^2) \\
 &= \langle \mathbf{u}, \mathbf{w} \rangle + \langle \mathbf{v}, \mathbf{w} \rangle + 2(u_1 v_1 w_1^2 + u_2 v_2 w_2^2 + u_3 v_3 w_3^2) \\
 &\neq \langle \mathbf{u}, \mathbf{w} \rangle + \langle \mathbf{v}, \mathbf{w} \rangle
 \end{aligned}$$

Additivity axiom is failed.

Hence, the given product is not an inner product in R^3 .

Example 4: If $\mathbf{u} = (u_1, u_2)$, $\mathbf{v} = (v_1, v_2)$ are vectors in R^2 then prove that R^2 is an inner product space with respect to the inner product defined as

$$\langle \mathbf{u}, \mathbf{v} \rangle = 4u_1v_1 + u_2v_1 + 4u_1v_2 + 4u_2v_2.$$

4.2.3 Norm or Length in Inner Product Spaces

The norm or length of a vector $\mathbf{u}$ in an inner product space V is denoted by $\|\mathbf{u}\|$ and is defined by

$$\|\mathbf{u}\| = \langle \mathbf{u}, \mathbf{u} \rangle^{\frac{1}{2}}$$

Unit Vector: Let $\mathbf{u}$ be a vector in an inner product space V . If $\|\mathbf{u}\| = 1$ then $\mathbf{u}$ is called a unit vector in V .

Properties of Length

If $\mathbf{u}$ and $\mathbf{v}$ are vectors in an inner product space V and k is any scalar then

1. $\|\mathbf{u}\| \geq 0$
2. $\|\mathbf{u}\| = 0$ if and only if $\mathbf{u} = 0$
3. $\|k\mathbf{u}\| = |k| \|\mathbf{u}\|$
4. $\|\mathbf{u} + \mathbf{v}\| \leq \|\mathbf{u}\| + \|\mathbf{v}\|$, Triangle inequality.

4.2.4 *Distance in Inner Product Spaces*

The distance between two vectors $\mathbf{u}$ and $\mathbf{v}$ in an inner product space V is denoted by $d(\mathbf{u}, \mathbf{v})$ and is defined by

$$d(\mathbf{u}, \mathbf{v}) = \|\mathbf{u} - \mathbf{v}\|$$

Properties of Distance

If $\mathbf{u}$, $\mathbf{v}$ and $\mathbf{w}$ are vectors in an inner product space V then

1. $d(\mathbf{u}, \mathbf{v}) \geq 0$
2. $d(\mathbf{u}, \mathbf{v}) = 0$ if and only if $\mathbf{u} = \mathbf{v}$
3. $d(\mathbf{u}, \mathbf{v}) = d(\mathbf{v}, \mathbf{u})$
4. $d(\mathbf{u}, \mathbf{v}) \leq d(\mathbf{u}, \mathbf{w}) + d(\mathbf{w}, \mathbf{v})$, Triangle inequality

Example 1: Find $\|\mathbf{u}\|$ if $\mathbf{u} = (3, 4)$ and weighted Euclidean inner product is $\langle \mathbf{u}, \mathbf{v} \rangle = u_1 v_1 - u_1 v_2 - u_2 v_1 + 3u_2 v_2$ where $\mathbf{u} = (u_1, u_2)$, $\mathbf{v} = (v_1, v_2)$.

Solution:

$$\begin{aligned}\|\mathbf{u}\| &= \langle \mathbf{u}, \mathbf{u} \rangle^{\frac{1}{2}} \\ &= (u_1 u_1 - u_1 u_2 - u_2 u_1 + 3u_2 u_2)^{\frac{1}{2}} \\ &= (3^2 - 3 \cdot 4 - 4 \cdot 3 + 3(4)^2)^{\frac{1}{2}} \\ &= \sqrt{33}.\end{aligned}$$

Example 7: Find $\langle \mathbf{f}, \mathbf{g} \rangle$ if $\mathbf{f} = f(x) = 1 - x + x^2 + 5x^3$, $\mathbf{g} = g(x) = x - 3x^2$ and the inner product $\langle \mathbf{f}, \mathbf{g} \rangle = \int_{-1}^1 f(x)g(x) dx$

Solution: $\langle \mathbf{f}, \mathbf{g} \rangle = \int_{-1}^1 (1 - x + x^2 + 5x^3)(x - 3x^2) dx$

$$= \int_{-1}^1 (x - 3x^2 - x^2 + 3x^3 + x^3 - 3x^4 + 5x^4 - 15x^5) dx$$

$$= \int_{-1}^1 (x - 4x^2 + 4x^3 + 2x^4 - 15x^5) dx$$

$$= 2 \int_0^1 (-4x^2 + 2x^4) dx \left[\begin{array}{l} \because \int_{-a}^a f(x) dx = 0, \text{ if } f(x) \text{ is odd} \\ = 2 \int_0^a f(x) dx, \text{ if } f(x) \text{ is even} \end{array} \right]$$

$$= 2 \left| -4 \frac{x^3}{3} + 2 \frac{x^5}{5} \right|_0^1$$

$$= 2 \left(-\frac{4}{3} + \frac{2}{5} \right)$$

$$= -\frac{28}{15}.$$

Example 8: Find $d(\mathbf{f}, \mathbf{g})$ if $\mathbf{f} = f(x) = \cos 2\pi x$ and $\mathbf{g} = g(x) = \sin 2\pi x$ and the inner product $\langle \mathbf{f}, \mathbf{g} \rangle = \int_0^1 f(x)g(x) dx$

Solution: $d(\mathbf{f}, \mathbf{g}) = \|\mathbf{f} - \mathbf{g}\|$

$$= \langle \mathbf{f} - \mathbf{g}, \mathbf{f} - \mathbf{g} \rangle^{\frac{1}{2}}$$

$$\langle \mathbf{f} - \mathbf{g}, \mathbf{f} - \mathbf{g} \rangle = \int_0^1 [f(x) - g(x)][f(x) - g(x)] dx$$

$$= \int_0^1 (\cos 2\pi x - \sin 2\pi x)^2 dx$$

$$= \int_0^1 (\cos^2 2\pi x + \sin^2 2\pi x - 2 \cos 2\pi x \sin 2\pi x) dx$$

$$= \int_0^1 (1 - \sin 4\pi x) dx$$

$$= \left| x + \frac{\cos 4\pi x}{4\pi} \right|_0^1$$

$$= 1 + \frac{\cos 4\pi - \cos 0}{4\pi}$$

$$= 1$$

$$d(\mathbf{f}, \mathbf{g}) = \sqrt{1} = 1.$$

4.2.5 Angle between Vectors

If $\mathbf{u}$ and $\mathbf{v}$ are non-zero vectors in an inner product space V and if θ is the angle between them then

$$\cos \theta = \frac{\langle \mathbf{u}, \mathbf{v} \rangle}{\|\mathbf{u}\| \|\mathbf{v}\|}$$

4.2.6 Orthogonality

Two vectors $\mathbf{u}$ and $\mathbf{v}$ in an inner product space V are called orthogonal if

$$\langle \mathbf{u}, \mathbf{v} \rangle = 0 \quad \text{i.e.,} \quad \theta = \frac{\pi}{2}$$

4.2.7 Pythagorean Theorem

If $\mathbf{u}$ and $\mathbf{v}$ are orthogonal vectors in an inner product space V then

$$\|\mathbf{u} + \mathbf{v}\|^2 = \|\mathbf{u}\|^2 + \|\mathbf{v}\|^2$$

Proof: Since $\mathbf{u}$ and $\mathbf{v}$ are orthogonal, $\langle \mathbf{u}, \mathbf{v} \rangle = \langle \mathbf{v}, \mathbf{u} \rangle = 0$

$$\begin{aligned} \|\mathbf{u} + \mathbf{v}\|^2 &= \langle \mathbf{u} + \mathbf{v}, \mathbf{u} + \mathbf{v} \rangle \\ &= \langle \mathbf{u}, \mathbf{u} + \mathbf{v} \rangle + \langle \mathbf{v}, \mathbf{u} + \mathbf{v} \rangle \\ &= \langle \mathbf{u}, \mathbf{u} \rangle + \langle \mathbf{u}, \mathbf{v} \rangle + \langle \mathbf{v}, \mathbf{u} \rangle + \langle \mathbf{v}, \mathbf{v} \rangle \\ &= \|\mathbf{u}\|^2 + \|\mathbf{v}\|^2 \quad \left[\because \langle \mathbf{u}, \mathbf{v} \rangle = \langle \mathbf{v}, \mathbf{u} \rangle = 0 \right] \end{aligned}$$

4.2.8 Cauchy–Schwarz Inequality

If $\mathbf{u}$ and $\mathbf{v}$ are vectors in an inner product space V then

$$|\langle \mathbf{u}, \mathbf{v} \rangle| \leq \|\mathbf{u}\| \|\mathbf{v}\|$$

Example 1: Find the cosine of the angle between $\mathbf{u}$ and $\mathbf{v}$ if R^2 , R^3 and R^4 have the Euclidean inner product.

- (i) $\mathbf{u} = (1, -3), \mathbf{v} = (2, 4)$
- (ii) $\mathbf{u} = (-1, 5, 2), \mathbf{v} = (2, 4, -9)$
- (iii) $\mathbf{u} = (1, 0, 1, 0), \mathbf{v} = (-3, -3, -3, -3)$

Solution: $\cos \theta = \frac{\langle \mathbf{u}, \mathbf{v} \rangle}{\|\mathbf{u}\| \|\mathbf{v}\|}$, where θ is angle between $\mathbf{u}$ & $\mathbf{v}$.

$$= \frac{\mathbf{u} \cdot \mathbf{v}}{\|\mathbf{u}\| \|\mathbf{v}\|}$$

$$\begin{aligned} \text{(i) } \cos \theta &= \frac{(1, -3) \cdot (2, 4)}{\sqrt{1+9} \sqrt{4+16}} \\ &= \frac{2-12}{\sqrt{10} \sqrt{20}} \\ &= \frac{-10}{\sqrt{10} \sqrt{20}} \\ &= -\frac{1}{\sqrt{2}} \end{aligned}$$

$$\begin{aligned} \text{(ii) } \cos \theta &= \frac{(-1, 5, 2) \cdot (2, 4, -9)}{\sqrt{1+25+4} \sqrt{4+16+81}} \\ &= \frac{-2+20-18}{\sqrt{30} \sqrt{101}} \\ &= 0 \end{aligned}$$

Example 4: Find cosine of the angle between $\mathbf{p}_1 = x - x^2$, and $\mathbf{p}_2 = 7 + 3x + 3x^2$ if the inner product $\langle \mathbf{p}_1, \mathbf{p}_2 \rangle = a_0b_0 + a_1b_1 + a_2b_2$, where

$$\mathbf{p}_1 = a_0 + a_1x + a_2x^2 \text{ and } \mathbf{p}_2 = b_0 + b_1x + b_2x^2.$$

Solution: Let θ be the angle between $\mathbf{p}_1$ and $\mathbf{p}_2$.

$$\begin{aligned}\cos \theta &= \frac{\langle \mathbf{p}_1, \mathbf{p}_2 \rangle}{\|\mathbf{p}_1\| \|\mathbf{p}_2\|} \\&= \frac{\langle x - x^2, 7 + 3x + 3x^2 \rangle}{\sqrt{(1)^2 + (-1)^2} \sqrt{(7)^2 + (3)^2 + (3)^2}} \\&= \frac{(0)(7) + (1)(3) + (-1)(3)}{\sqrt{2} \sqrt{67}} \\&= 0\end{aligned}$$

Example 2: Determine whether the given vectors are orthogonal with respect to the Euclidean inner product.

- (i) $\mathbf{u} = (-1, 3, 2), \mathbf{v} = (4, 2, -1)$
- (ii) $\mathbf{u} = (-4, 6, -10, 1), \mathbf{v} = (2, 1, -2, 9)$

Solution: (i) $\langle \mathbf{u}, \mathbf{v} \rangle = \mathbf{u} \cdot \mathbf{v}$

$$\begin{aligned} &= (-1, 3, 2) \cdot (4, 2, -1) \\ &= (-1)(4) + (3)(2) + (2)(-1) \\ &= -4 + 6 - 2 \\ &= 0 \end{aligned}$$

Hence, $\mathbf{u}$ and $\mathbf{v}$ are orthogonal.

(ii) $\langle \mathbf{u}, \mathbf{v} \rangle = \mathbf{u} \cdot \mathbf{v}$

$$\begin{aligned} &= (-4, 6, -10, 1) \cdot (2, 1, -2, 9) \\ &= (-4)(2) + (6)(1) + (-10)(-2) + (1)(9) \\ &= -8 + 6 + 20 + 9 \\ &= 27 \neq 0 \end{aligned}$$

Hence, $\mathbf{u}$ and $\mathbf{v}$ are not orthogonal.

Example 3: Determine whether there exists scalars k and l such that the vectors $\mathbf{u} = (2, k, 6)$, $\mathbf{v} = (l, 5, 3)$ and $\mathbf{w} = (1, 2, 3)$ are mutually orthogonal with respect to the Euclidean inner product.

Solution: Let $\mathbf{u}$, $\mathbf{v}$ and $\mathbf{w}$ be mutually orthogonal.

$$\begin{aligned}\langle \mathbf{u}, \mathbf{v} \rangle &= \mathbf{u} \cdot \mathbf{v} = 0 \\ (2, k, 6) \cdot (l, 5, 3) &= 0 \\ 2l + 5k + 18 &= 0 \\ 2l + 5k &= -18 \quad \dots(1)\end{aligned}$$

$$\begin{aligned}\langle \mathbf{v}, \mathbf{w} \rangle &= \mathbf{v} \cdot \mathbf{w} = 0 \\ (l, 5, 3) \cdot (1, 2, 3) &= 0 \\ l + 10 + 9 &= 0 \\ l &= -19 \quad \dots(2)\end{aligned}$$

$$\begin{aligned}\langle \mathbf{w}, \mathbf{u} \rangle &= \mathbf{w} \cdot \mathbf{u} = 0 \\ (1, 2, 3) \cdot (2, k, 6) &= 0 \\ 2 + 2k + 18 &= 0 \\ k &= -10 \quad \dots(3)\end{aligned}$$

Substituting l and k in equation (1),

$$\begin{aligned}2(-19) + 5(-10) &= -18 \\ -88 &= -18\end{aligned}$$

This shows that l and k does not satisfy equation (1).

Hence, there do not exist k and l such that $\mathbf{u}$, $\mathbf{v}$ and $\mathbf{w}$ are orthogonal.

Example 5: Show that $\mathbf{p}_1 = 1 - x + 2x^2$ and $\mathbf{p}_2 = 2x + x^2$ are orthogonal with respect to the inner product $\langle \mathbf{p}_1, \mathbf{p}_2 \rangle = a_0b_0 + a_1b_1 + a_2b_2$, where

$$\mathbf{p}_1 = a_0 + a_1x + a_2x^2 \text{ and } \mathbf{p}_2 = b_0 + b_1x + b_2x^2.$$

Solution:

$$\begin{aligned}\langle \mathbf{p}_1, \mathbf{p}_2 \rangle &= \langle 1 - x + 2x^2, 2x + x^2 \rangle \\ &= (1)(0) + (-1)(2) + (2)(1) \\ &= 0\end{aligned}$$

Hence, $\mathbf{p}_1$ and $\mathbf{p}_2$ are orthogonal.

Example 6: Show that the matrices $A = \begin{bmatrix} 2 & 1 \\ -1 & 3 \end{bmatrix}$ and $B = \begin{bmatrix} 1 & 1 \\ 0 & -1 \end{bmatrix}$ are orthogonal with respect to the inner product $\langle A, B \rangle = a_1b_1 + a_2b_2 + a_3b_3 + a_4b_4$ where

$$A = \begin{bmatrix} a_1 & a_2 \\ a_3 & a_4 \end{bmatrix} \text{ and } B = \begin{bmatrix} b_1 & b_2 \\ b_3 & b_4 \end{bmatrix}.$$

Solution:

$$\begin{aligned}\langle A, B \rangle &= (2)(1) + (1)(1) + (-1)(0) + (3)(-1) \\ &= 0\end{aligned}$$

Hence, A and B are orthogonal.

4.2.9 Orthogonal and Orthonormal Set

A set $S = \{\mathbf{u}_1, \mathbf{u}_2, \dots, \mathbf{u}_p\}$ of vectors in an inner product space V is called an orthogonal set if each pair of distinct vectors in S are orthogonal, i.e. $\langle \mathbf{u}_i, \mathbf{u}_j \rangle = 0$ for $i \neq j$.

An orthogonal set of unit vectors (norm is 1) is called orthonormal, i.e. $\langle \mathbf{u}_i, \mathbf{u}_j \rangle = 0$ for $i \neq j$ and $\langle \mathbf{u}_i, \mathbf{u}_i \rangle = 1$ for $i = 1, 2, \dots, p$.

The process of dividing a non-zero vector $\mathbf{u}$ by its norm is called normalizing $\mathbf{u}$.

$$\text{normalized } \mathbf{u} = \frac{\mathbf{u}}{\|\mathbf{u}\|}$$

Example 1: Show that the vectors $\mathbf{u}_1 = \left(\frac{2}{3}, -\frac{2}{3}, \frac{1}{3}\right)$, $\mathbf{u}_2 = \left(\frac{2}{3}, \frac{1}{3}, -\frac{2}{3}\right)$ and $\mathbf{u}_3 = \left(\frac{1}{3}, \frac{2}{3}, \frac{2}{3}\right)$ are orthonormal with respect to the Euclidean inner product on R^3 .

Solution:

$$\begin{aligned}\langle \mathbf{u}_1, \mathbf{u}_2 \rangle &= \left(\frac{2}{3}, -\frac{2}{3}, \frac{1}{3}\right) \cdot \left(\frac{2}{3}, \frac{1}{3}, -\frac{2}{3}\right) \\ &= \left(\frac{2}{3}\right)\left(\frac{2}{3}\right) + \left(-\frac{2}{3}\right)\left(\frac{1}{3}\right) + \frac{1}{3}\left(-\frac{2}{3}\right) \\ &= 0\end{aligned}$$

$$\begin{aligned}
\langle \mathbf{u}_2, \mathbf{u}_3 \rangle &= \left(\frac{2}{3}, \frac{1}{3}, -\frac{2}{3} \right) \cdot \left(\frac{1}{3}, \frac{2}{3}, \frac{2}{3} \right) \\
&= \left(\frac{2}{3} \right) \left(\frac{1}{3} \right) + \left(\frac{1}{3} \right) \left(\frac{2}{3} \right) + \left(-\frac{2}{3} \right) \left(\frac{2}{3} \right) \\
&= 0 \\
\langle \mathbf{u}_3, \mathbf{u}_1 \rangle &= \left(\frac{1}{3}, \frac{2}{3}, \frac{2}{3} \right) \cdot \left(\frac{2}{3}, -\frac{2}{3}, \frac{1}{3} \right) \\
&= \left(\frac{1}{3} \right) \left(\frac{2}{3} \right) + \left(\frac{2}{3} \right) \left(-\frac{2}{3} \right) + \left(\frac{2}{3} \right) \left(\frac{1}{3} \right) \\
&= 0
\end{aligned}$$

$$\|\mathbf{u}_1\| = \langle \mathbf{u}_1, \mathbf{u}_1 \rangle^{\frac{1}{2}} = \left[\left(\frac{2}{3} \right)^2 + \left(-\frac{2}{3} \right)^2 + \left(\frac{1}{3} \right)^2 \right]^{\frac{1}{2}} = 1$$

$$\|\mathbf{u}_2\| = \langle \mathbf{u}_2, \mathbf{u}_2 \rangle^{\frac{1}{2}} = \left[\left(\frac{2}{3} \right)^2 + \left(\frac{1}{3} \right)^2 + \left(-\frac{2}{3} \right)^2 \right]^{\frac{1}{2}} = 1$$

$$\|\mathbf{u}_3\| = \langle \mathbf{u}_3, \mathbf{u}_3 \rangle^{\frac{1}{2}} = \left[\left(\frac{1}{3} \right)^2 + \left(\frac{2}{3} \right)^2 + \left(\frac{2}{3} \right)^2 \right]^{\frac{1}{2}} = 1$$

Since $\langle \mathbf{u}_1, \mathbf{u}_2 \rangle = \langle \mathbf{u}_2, \mathbf{u}_3 \rangle = \langle \mathbf{u}_3, \mathbf{u}_1 \rangle = 0$ and $\|\mathbf{u}_1\| = \|\mathbf{u}_2\| = \|\mathbf{u}_3\| = 1$, the vectors are orthonormal.

Example 2: Show that the set of vectors $\mathbf{u}_1 = \left(\frac{1}{5}, \frac{1}{5}, \frac{1}{5}\right)$, $\mathbf{u}_2 = \left(-\frac{1}{2}, \frac{1}{2}, 0\right)$, $\mathbf{u}_3 = \left(\frac{1}{3}, \frac{1}{3}, -\frac{2}{3}\right)$ is orthogonal with respect to the Euclidean inner product on R^3 and then convert it to an orthonormal set by normalizing the vectors.

Solution:

$$\begin{aligned}\langle \mathbf{u}_1, \mathbf{u}_2 \rangle &= \left(\frac{1}{5}, \frac{1}{5}, \frac{1}{5}\right) \cdot \left(-\frac{1}{2}, \frac{1}{2}, 0\right) \\ &= \left(\frac{1}{5}\right)\left(-\frac{1}{2}\right) + \left(\frac{1}{5}\right)\left(\frac{1}{2}\right) + \left(\frac{1}{5}\right)(0) \\ &= 0\end{aligned}$$
$$\begin{aligned}\langle \mathbf{u}_2, \mathbf{u}_3 \rangle &= \left(-\frac{1}{2}, \frac{1}{2}, 0\right) \cdot \left(\frac{1}{3}, \frac{1}{3}, -\frac{2}{3}\right) \\ &= \left(-\frac{1}{2}\right)\left(\frac{1}{3}\right) + \left(\frac{1}{2}\right)\left(\frac{1}{3}\right) + (0)\left(-\frac{2}{3}\right) \\ &= 0\end{aligned}$$
$$\begin{aligned}\langle \mathbf{u}_3, \mathbf{u}_1 \rangle &= \left(\frac{1}{3}, \frac{1}{3}, -\frac{2}{3}\right) \cdot \left(\frac{1}{5}, \frac{1}{5}, \frac{1}{5}\right) \\ &= \left(\frac{1}{3}\right)\left(\frac{1}{5}\right) + \left(\frac{1}{3}\right)\left(\frac{1}{5}\right) + \left(-\frac{2}{3}\right)\left(\frac{1}{5}\right) \\ &= 0\end{aligned}$$

$$\|\mathbf{u}_1\| = \sqrt{\left(\frac{1}{5}\right)^2 + \left(\frac{1}{5}\right)^2 + \left(\frac{1}{5}\right)^2} = \sqrt{\frac{3}{25}} = \frac{\sqrt{3}}{5}$$

$$\|\mathbf{u}_2\| = \sqrt{\left(-\frac{1}{2}\right)^2 + \left(\frac{1}{2}\right)^2} = \sqrt{\frac{1}{2}}$$

$$\|\mathbf{u}_3\| = \sqrt{\left(\frac{1}{3}\right)^2 + \left(\frac{1}{3}\right)^2 + \left(-\frac{2}{3}\right)^2} = \sqrt{\frac{2}{3}}$$

Normalising the vectors,

$$\mathbf{w}_1 = \frac{\mathbf{u}_1}{\|\mathbf{u}_1\|} = \left(\frac{1}{\sqrt{3}}, \frac{1}{\sqrt{3}}, \frac{1}{\sqrt{3}} \right)$$

$$\mathbf{w}_2 = \frac{\mathbf{u}_2}{\|\mathbf{u}_2\|} = \left(-\frac{1}{\sqrt{2}}, \frac{1}{\sqrt{2}}, 0 \right)$$

$$\mathbf{w}_3 = \frac{\mathbf{u}_3}{\|\mathbf{u}_3\|} = \left(\frac{1}{\sqrt{6}}, \frac{1}{\sqrt{6}}, -\sqrt{\frac{2}{3}} \right)$$

Orthonormal set = $\{\mathbf{w}_1, \mathbf{w}_2, \mathbf{w}_3\}$

EXAMPLE 7.7 Find a nonzero vector w that is orthogonal to $u_1 = (1, 2, 1)$ and $u_2 = (2, 5, 4)$ in $\mathbb{R}^3$.

Let $w = (x, y, z)$. Then we want $\langle u_1, w \rangle = 0$ and $\langle u_2, w \rangle = 0$. This yields the homogeneous system

$$\begin{array}{rcl} x + 2y + z = 0 & \text{or} & x + 2y + z = 0 \\ 2x + 5y + 4z = 0 & & y + 2z = 0 \end{array}$$

Here z is the only free variable in the echelon system. Set $z = 1$ to obtain $y = -2$ and $x = 3$. Thus, $w = (3, -2, 1)$ is a desired nonzero vector orthogonal to u_1 and u_2 .

Any multiple of w will also be orthogonal to u_1 and u_2 . Normalizing w , we obtain the following unit vector orthogonal to u_1 and u_2 :

$$\hat{w} = \frac{w}{\|w\|} = \left(\frac{3}{\sqrt{14}}, -\frac{2}{\sqrt{14}}, \frac{1}{\sqrt{14}} \right)$$

EXAMPLE 7.9

- (a) Let $E = \{e_1, e_2, e_3\} = \{(1, 0, 0), (0, 1, 0), (0, 0, 1)\}$ be the usual basis of Euclidean space $\mathbf{R}^3$. It is clear that

$$\langle e_1, e_2 \rangle = \langle e_1, e_3 \rangle = \langle e_2, e_3 \rangle = 0 \quad \text{and} \quad \langle e_1, e_1 \rangle = \langle e_2, e_2 \rangle = \langle e_3, e_3 \rangle = 1$$

Namely, E is an orthonormal basis of $\mathbf{R}^3$. More generally, the usual basis of $\mathbf{R}^n$ is orthonormal for every n .

- (b) Let $V = C[-\pi, \pi]$ be the vector space of continuous functions on the interval $-\pi \leq t \leq \pi$ with inner product defined by $\langle f, g \rangle = \int_{-\pi}^{\pi} f(t)g(t) dt$. Then the following is a classical example of an orthogonal set in V :

$$\{1, \cos t, \cos 2t, \cos 3t, \dots, \sin t, \sin 2t, \sin 3t, \dots\}$$

Orthogonal Complements

Let S be a subset of an inner product space V . The orthogonal complement of S , denoted by $S^\perp$ (read “ S perp”) consists of those vectors in V that are orthogonal to every vector $u \in S$; that is,

$$S^\perp = \{v \in V : \langle v, u \rangle = 0 \text{ for every } u \in S\}$$

In particular, for a given vector u in V , we have

$$u^\perp = \{v \in V : \langle v, u \rangle = 0\}$$

that is, $u^\perp$ consists of all vectors in V that are orthogonal to the given vector u .

EXAMPLE 7.8 Find a basis for the subspace $u^\perp$ of $\mathbf{R}^3$, where $u = (1, 3, -4)$.

Note that $u^\perp$ consists of all vectors $w = (x, y, z)$ such that $\langle u, w \rangle = 0$, or $x + 3y - 4z = 0$. The free variables are y and z .

- (1) Set $y = 1, z = 0$ to obtain the solution $w_1 = (-3, 1, 0)$.
- (2) Set $y = 0, z = 1$ to obtain the solution $w_2 = (4, 0, 1)$.

THEOREM 7.4: Let W be a subspace of V . Then V is the direct sum of W and $W^\perp$; that is, $V = W \oplus W^\perp$.

4.3 ORTHOGONAL AND ORTHONORMAL BASIS

In an inner product space, a basis consisting of orthogonal vectors is called an orthogonal basis.

In an inner product space, a basis consisting of orthonormal vectors is called an orthonormal basis.

Theorem 4.1: If $S = \{\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_n\}$ is an orthogonal set of non-zero vectors in an inner product space V then S is linearly independent.

Theorem 4.2: Any orthogonal set of n non-zero vectors in R^n is a basis for R^n .

Theorem 4.3: If $S = \{\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_n\}$ is an orthonormal basis for an inner product space V , and $\mathbf{u}$ is any vector in V then it can be expressed as a linear combination of $\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_n$.

$$\mathbf{u} = \langle \mathbf{u}, \mathbf{v}_1 \rangle \mathbf{v}_1 + \langle \mathbf{u}, \mathbf{v}_2 \rangle \mathbf{v}_2 + \dots + \langle \mathbf{u}, \mathbf{v}_n \rangle \mathbf{v}_n$$

Here, $\langle \mathbf{u}, \mathbf{v}_1 \rangle, \langle \mathbf{u}, \mathbf{v}_2 \rangle, \dots, \langle \mathbf{u}, \mathbf{v}_n \rangle$ are the coordinate vectors of $\mathbf{u}$ with respect to the orthonormal basis S ,

i.e.,
$$[\mathbf{u}]_S = (\langle \mathbf{u}, \mathbf{v}_1 \rangle, \langle \mathbf{u}, \mathbf{v}_2 \rangle, \dots, \langle \mathbf{u}, \mathbf{v}_n \rangle)$$

Corollary: If $S = \{\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_n\}$ is an orthogonal basis for an inner product space V , and $\mathbf{u}$ is any vector in V then

$$\mathbf{u} = \frac{\langle \mathbf{u}, \mathbf{v}_1 \rangle}{\|\mathbf{v}_1\|^2} \mathbf{v}_1 + \frac{\langle \mathbf{u}, \mathbf{v}_2 \rangle}{\|\mathbf{v}_2\|^2} \mathbf{v}_2 + \dots + \frac{\langle \mathbf{u}, \mathbf{v}_n \rangle}{\|\mathbf{v}_n\|^2} \mathbf{v}_n$$

Theorem 4.4: If S is an orthonormal basis for an n -dimensional inner product space, and if coordinate vectors of $\mathbf{u}$ and $\mathbf{v}$ with respect to S are $[\mathbf{u}]_S = (a_1, a_2, \dots, a_n)$ and $[\mathbf{v}]_S = (b_1, b_2, \dots, b_n)$ then,

- (i) $\|\mathbf{u}\| = \sqrt{a_1^2 + a_2^2 + \dots + a_n^2}$
- (ii) $d(\mathbf{u}, \mathbf{v}) = \sqrt{(a_1 - b_1)^2 + (a_2 - b_2)^2 + \dots + (a_n - b_n)^2}$
- (iii) $\langle \mathbf{u}, \mathbf{v} \rangle = a_1 b_1 + a_2 b_2 + \dots + a_n b_n = [\mathbf{u}]_S \cdot [\mathbf{v}]_S$

Orthogonal Basis and Linear Combinations, Fourier Coefficients

Let S consist of the following three vectors in $\mathbf{R}^3$:

$$u_1 = (1, 2, 1), \quad u_2 = (2, 1, -4), \quad u_3 = (3, -2, 1)$$

The reader can verify that the vectors are orthogonal; hence, they are linearly independent. Thus, S is an orthogonal basis of $\mathbf{R}^3$.

Suppose we want to write $v = (7, 1, 9)$ as a linear combination of u_1, u_2, u_3 . First we set v as a linear combination of u_1, u_2, u_3 using unknowns x_1, x_2, x_3 as follows:

$$v = x_1 u_1 + x_2 u_2 + x_3 u_3 \quad \text{or} \quad (7, 1, 9) = x_1(1, 2, 1) + x_2(2, 1, -4) + x_3(3, -2, 1) \quad (*)$$

We can proceed in two ways.

METHOD 1: Expand $(*)$ (as in Chapter 3) to obtain the system

$$x_1 + 2x_2 + 3x_3 = 7, \quad 2x_1 + x_2 - 2x_3 = 1, \quad x_1 - 4x_2 + x_3 = 7$$

Solve the system by Gaussian elimination to obtain $x_1 = 3$, $x_2 = -1$, $x_3 = 2$. Thus, $v = 3u_1 - u_2 + 2u_3$.

METHOD 2: (This method uses the fact that the basis vectors are orthogonal, and the arithmetic is much simpler.) If we take the inner product of each side of (*) with respect to u_i , we get

$$\langle v, u_i \rangle = \langle x_1 u_1 + x_2 u_2 + x_3 u_3, u_i \rangle \quad \text{or} \quad \langle v, u_i \rangle = x_i \langle u_i, u_i \rangle \quad \text{or} \quad x_i = \frac{\langle v, u_i \rangle}{\langle u_i, u_i \rangle}$$

Here two terms drop out, because u_1, u_2, u_3 are orthogonal. Accordingly,

$$x_1 = \frac{\langle v, u_1 \rangle}{\langle u_1, u_1 \rangle} = \frac{7 + 2 + 9}{1 + 4 + 1} = \frac{18}{6} = 3, \quad x_2 = \frac{\langle v, u_2 \rangle}{\langle u_2, u_2 \rangle} = \frac{14 + 1 - 36}{4 + 1 + 16} = \frac{-21}{21} = -1$$

$$x_3 = \frac{\langle v, u_3 \rangle}{\langle u_3, u_3 \rangle} = \frac{21 - 2 + 9}{9 + 4 + 1} = \frac{28}{14} = 2$$

Thus, again, we get $v = 3u_1 - u_2 + 2u_3$.

THEOREM 7.7: Let $\{u_1, u_2, \dots, u_n\}$ be an orthogonal basis of V . Then, for any $v \in V$,

$$v = \frac{\langle v, u_1 \rangle}{\langle u_1, u_1 \rangle} u_1 + \frac{\langle v, u_2 \rangle}{\langle u_2, u_2 \rangle} u_2 + \dots + \frac{\langle v, u_n \rangle}{\langle u_n, u_n \rangle} u_n$$

Remark: The scalar $k_i \equiv \frac{\langle v, u_i \rangle}{\langle u_i, u_i \rangle}$ is called the *Fourier coefficient* of v with respect to u_i , because it is analogous to a coefficient in the Fourier series of a function. This scalar also has a geometric interpretation, which is discussed below.

Projections

Let V be an inner product space. Suppose w is a given nonzero vector in V , and suppose v is another vector. We seek the “projection of v along w ,” which, as indicated in Fig. 7-3(a), will be the multiple cw of w such that $v' = v - cw$ is orthogonal to w . This means

$$\langle v - cw, w \rangle = 0 \quad \text{or} \quad \langle v, w \rangle - c\langle w, w \rangle = 0 \quad \text{or} \quad c = \frac{\langle v, w \rangle}{\langle w, w \rangle}$$

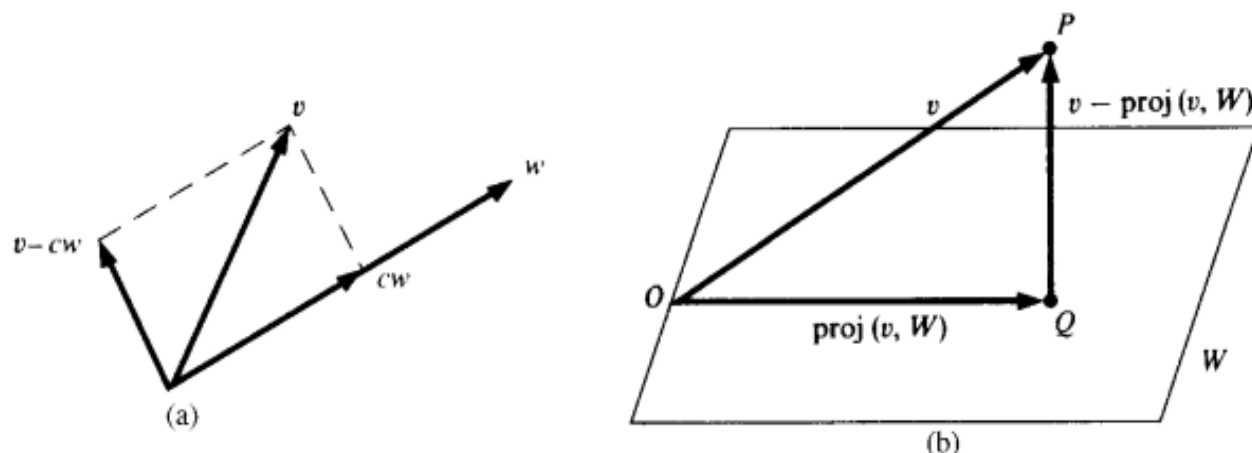


Figure 7-3

Accordingly, the *projection of v along w* is denoted and defined by

$$\text{proj}(v, w) = cw = \frac{\langle v, w \rangle}{\langle w, w \rangle} w$$

Such a scalar c is unique, and it is called the *Fourier coefficient* of v with respect to w or the *component* of v along w .

THEOREM 7.8: Suppose $w_1, w_2, \dots, w_r$ form an orthogonal set of nonzero vectors in V . Let v be any vector in V . Define

$$v' = v - (c_1 w_1 + c_2 w_2 + \cdots + c_r w_r)$$

where

$$c_1 = \frac{\langle v, w_1 \rangle}{\langle w_1, w_1 \rangle}, \quad c_2 = \frac{\langle v, w_2 \rangle}{\langle w_2, w_2 \rangle}, \quad \dots, \quad c_r = \frac{\langle v, w_r \rangle}{\langle w_r, w_r \rangle}$$

Then v' is orthogonal to $w_1, w_2, \dots, w_r$.

Note that each c_i in the above theorem is the component (Fourier coefficient) of v along the given w_i .

Remark: The notion of the projection of a vector $v \in V$ along a subspace W of V is defined as follows. By Theorem 7.4, $V = W \oplus W^\perp$. Hence, v may be expressed uniquely in the form

$$v = w + w', \quad \text{where} \quad w \in W \quad \text{and} \quad w' \in W^\perp$$

We define w to be the *projection of v along W* , and denote it by $\text{proj}(v, W)$, as pictured in Fig. 7-2(b). In particular, if $W = \text{span}(w_1, w_2, \dots, w_r)$, where the w_i form an orthogonal set, then

$$\text{proj}(v, W) = c_1 w_1 + c_2 w_2 + \cdots + c_r w_r$$

Here c_i is the component of v along w_i , as above.

Example 1: Find the orthogonal projection of $\mathbf{u} = (1, -2, 3)$ along $\mathbf{v} = (1, 2, 1)$ in R^3 with respect to the Euclidean inner product.

Solution: Let W be the subspace spanned by the vector $\mathbf{v} = (1, 2, 1)$.

$$\begin{aligned}\text{proj}_W \mathbf{u} &= \frac{\langle \mathbf{u}, \mathbf{v} \rangle}{\|\mathbf{v}\|^2} \mathbf{v} \\ &= \frac{\mathbf{u} \cdot \mathbf{v}}{\|\mathbf{v}\|^2} \mathbf{v} \\ &= \frac{(1, -2, 3) \cdot (1, 2, 1)}{(1^2 + 2^2 + 1^2)} (1, 2, 1) \\ &= \frac{1 - 4 + 3}{6} (1, 2, 1) \\ &= 0\end{aligned}$$

7.7 Gram–Schmidt Orthogonalization Process

Suppose $\{v_1, v_2, \dots, v_n\}$ is a basis of an inner product space V . One can use this basis to construct an orthogonal basis $\{w_1, w_2, \dots, w_n\}$ of V as follows. Set

$$w_1 = v_1$$

$$w_2 = v_2 - \frac{\langle v_2, w_1 \rangle}{\langle w_1, w_1 \rangle} w_1$$

$$w_3 = v_3 - \frac{\langle v_3, w_1 \rangle}{\langle w_1, w_1 \rangle} w_1 - \frac{\langle v_3, w_2 \rangle}{\langle w_2, w_2 \rangle} w_2$$

.....

$$w_n = v_n - \frac{\langle v_n, w_1 \rangle}{\langle w_1, w_1 \rangle} w_1 - \frac{\langle v_n, w_2 \rangle}{\langle w_2, w_2 \rangle} w_2 - \dots - \frac{\langle v_n, w_{n-1} \rangle}{\langle w_{n-1}, w_{n-1} \rangle} w_{n-1}$$

In other words, for $k = 2, 3, \dots, n$, we define

$$w_k = v_k - c_{k1}w_1 - c_{k2}w_2 - \dots - c_{k,k-1}w_{k-1}$$

where $c_{ki} = \langle v_k, w_i \rangle / \langle w_i, w_i \rangle$ is the component of v_k along w_i . By Theorem 7.8, each w_k is orthogonal to the preceding w 's. Thus, $w_1, w_2, \dots, w_n$ form an orthogonal basis for V as claimed. Normalizing each w_i will then yield an orthonormal basis for V .

The above construction is known as the *Gram–Schmidt orthogonalization process*. The following remarks are in order.

EXAMPLE 7.10 Apply the Gram–Schmidt orthogonalization process to find an orthogonal basis and then an orthonormal basis for the subspace U of $\mathbf{R}^4$ spanned by

$$v_1 = (1, 1, 1, 1), \quad v_2 = (1, 2, 4, 5), \quad v_3 = (1, -3, -4, -2)$$

(1) First set $w_1 = v_1 = (1, 1, 1, 1)$.

(2) Compute

$$v_2 - \frac{\langle v_2, w_1 \rangle}{\langle w_1, w_1 \rangle} w_1 = v_2 - \frac{12}{4} w_1 = (-2, -1, 1, 2)$$

Set $w_2 = (-2, -1, 1, 2)$.

(3) Compute

$$v_3 - \frac{\langle v_3, w_1 \rangle}{\langle w_1, w_1 \rangle} w_1 - \frac{\langle v_3, w_2 \rangle}{\langle w_2, w_2 \rangle} w_2 = v_3 - \frac{(-8)}{4} w_1 - \frac{(-7)}{10} w_2 = \left(\frac{8}{5}, -\frac{17}{10}, -\frac{13}{10}, \frac{7}{5}\right)$$

Clear fractions to obtain $w_3 = (-6, -17, -13, 14)$.

Thus, w_1, w_2, w_3 form an orthogonal basis for U . Normalize these vectors to obtain an orthonormal basis $\{u_1, u_2, u_3\}$ of U . We have $\|w_1\|^2 = 4$, $\|w_2\|^2 = 10$, $\|w_3\|^2 = 910$, so

$$u_1 = \frac{1}{2}(1, 1, 1, 1), \quad u_2 = \frac{1}{\sqrt{10}}(-2, -1, 1, 2), \quad u_3 = \frac{1}{\sqrt{910}}(16, -17, -13, 14)$$

EXAMPLE 7.11 Let V be the vector space of polynomials $f(t)$ with inner product $\langle f, g \rangle = \int_{-1}^1 f(t)g(t) dt$. Apply the Gram–Schmidt orthogonalization process to $\{1, t, t^2, t^3\}$ to find an orthogonal basis $\{f_0, f_1, f_2, f_3\}$ with integer coefficients for $\mathbf{P}_3(t)$.

Here we use the fact that, for $r + s = n$,

$$\langle t^r, t^s \rangle = \int_{-1}^1 t^n dt = \frac{t^{n+1}}{n+1} \Big|_{-1}^1 = \begin{cases} 2/(n+1) & \text{when } n \text{ is even} \\ 0 & \text{when } n \text{ is odd} \end{cases}$$

- (1) First set $f_0 = 1$.
- (2) Compute $t = \frac{\langle t, 1 \rangle}{\langle 1, 1 \rangle} (1) = t - 0 = t$. Set $f_1 = t$.
- (3) Compute

$$t^2 - \frac{\langle t^2, 1 \rangle}{\langle 1, 1 \rangle} (1) - \frac{\langle t^2, t \rangle}{\langle t, t \rangle} (t) = t^2 - \frac{2}{2}(1) + 0(t) = t^2 - \frac{1}{3}$$

Multiply by 3 to obtain $f_2 = 3t^2 - 1$.

(4) Compute

$$\begin{aligned} t^3 - \frac{\langle t^3, 1 \rangle}{\langle 1, 1 \rangle} (1) - \frac{\langle t^3, t \rangle}{\langle t, t \rangle} (t) - \frac{\langle t^3, 3t^2 - 1 \rangle}{\langle 3t^2 - 1, 3t^2 - 1 \rangle} (3t^2 - 1) \\ = t^3 - 0(1) - \frac{2}{\frac{5}{3}}(t) - 0(3t^2 - 1) = t^3 - \frac{3}{5}t \end{aligned}$$

Multiply by 5 to obtain $f_3 = 5t^3 - 3t$.

Thus, $\{1, t, 3t^2 - 1, 5t^3 - 3t\}$ is the required orthogonal basis.

Remark: Normalizing the polynomials in Example 7.11 so that $p(1) = 1$ yields the polynomials

$$1, t, \frac{1}{2}(3t^2 - 1), \frac{1}{2}(5t^3 - 3t)$$

These are the first four *Legendre polynomials*, which appear in the study of differential equations.

► **EXAMPLE 8 Using the Gram–Schmidt Process**

Assume that the vector space R^3 has the Euclidean inner product. Apply the Gram–Schmidt process to transform the basis vectors

$$\mathbf{u}_1 = (1, 1, 1), \quad \mathbf{u}_2 = (0, 1, 1), \quad \mathbf{u}_3 = (0, 0, 1)$$

into an orthogonal basis $\{\mathbf{v}_1, \mathbf{v}_2, \mathbf{v}_3\}$, and then normalize the orthogonal basis vectors to obtain an orthonormal basis $\{\mathbf{q}_1, \mathbf{q}_2, \mathbf{q}_3\}$.

Solution

Step 1. $\mathbf{v}_1 = \mathbf{u}_1 = (1, 1, 1)$

$$\begin{aligned} \text{Step 2. } \mathbf{v}_2 &= \mathbf{u}_2 - \text{proj}_{W_1} \mathbf{u}_2 = \mathbf{u}_2 - \frac{\langle \mathbf{u}_2, \mathbf{v}_1 \rangle}{\|\mathbf{v}_1\|^2} \mathbf{v}_1 \\ &= (0, 1, 1) - \frac{2}{3}(1, 1, 1) = \left(-\frac{2}{3}, \frac{1}{3}, \frac{1}{3}\right) \end{aligned}$$

$$\begin{aligned} \text{Step 3. } \mathbf{v}_3 &= \mathbf{u}_3 - \text{proj}_{W_2} \mathbf{u}_3 = \mathbf{u}_3 - \frac{\langle \mathbf{u}_3, \mathbf{v}_1 \rangle}{\|\mathbf{v}_1\|^2} \mathbf{v}_1 - \frac{\langle \mathbf{u}_3, \mathbf{v}_2 \rangle}{\|\mathbf{v}_2\|^2} \mathbf{v}_2 \\ &= (0, 0, 1) - \frac{1}{3}(1, 1, 1) - \frac{1/3}{2/3} \left(-\frac{2}{3}, \frac{1}{3}, \frac{1}{3}\right) \\ &= \left(0, -\frac{1}{2}, \frac{1}{2}\right) \end{aligned}$$

$$\mathbf{v}_1 = (1, 1, 1), \quad \mathbf{v}_2 = \left(-\frac{2}{3}, \frac{1}{3}, \frac{1}{3}\right), \quad \mathbf{v}_3 = \left(0, -\frac{1}{2}, \frac{1}{2}\right)$$

form an orthogonal basis for R^3 . The norms of these vectors are

$$\|\mathbf{v}_1\| = \sqrt{3}, \quad \|\mathbf{v}_2\| = \frac{\sqrt{6}}{3}, \quad \|\mathbf{v}_3\| = \frac{1}{\sqrt{2}}$$

so an orthonormal basis for R^3 is

$$\mathbf{q}_1 = \frac{\mathbf{v}_1}{\|\mathbf{v}_1\|} = \left(\frac{1}{\sqrt{3}}, \frac{1}{\sqrt{3}}, \frac{1}{\sqrt{3}}\right), \quad \mathbf{q}_2 = \frac{\mathbf{v}_2}{\|\mathbf{v}_2\|} = \left(-\frac{2}{\sqrt{6}}, \frac{1}{\sqrt{6}}, \frac{1}{\sqrt{6}}\right),$$

$$\mathbf{q}_3 = \frac{\mathbf{v}_3}{\|\mathbf{v}_3\|} = \left(0, -\frac{1}{\sqrt{2}}, \frac{1}{\sqrt{2}}\right) \quad \blacktriangleleft$$

Matrix Representation of an Inner Product

Let $\mathcal{B} = \{\mathbf{u}_1, \mathbf{u}_2, \dots, \mathbf{u}_n\}$ be a basis of an n -dimensional inner product space V . For vectors $\mathbf{u}, \mathbf{v} \in V$, write

$$\mathbf{u} = x_1\mathbf{u}_1 + x_2\mathbf{u}_2 + \cdots + x_n\mathbf{u}_n,$$

$$\mathbf{v} = y_1\mathbf{u}_1 + y_2\mathbf{u}_2 + \cdots + y_n\mathbf{u}_n.$$

The linearity implies

$$\begin{aligned}\langle \mathbf{u}, \mathbf{v} \rangle &= \left\langle \sum_{i=1}^n x_i \mathbf{u}_i, \sum_{j=1}^n y_j \mathbf{u}_j \right\rangle \\ &= \sum_{i=1}^n \sum_{j=1}^n x_i y_j \langle \mathbf{u}_i, \mathbf{u}_j \rangle.\end{aligned}$$

We call the $n \times n$ matrix

$$A = \begin{bmatrix} \langle \mathbf{u}_1, \mathbf{u}_1 \rangle & \langle \mathbf{u}_1, \mathbf{u}_2 \rangle & \cdots & \langle \mathbf{u}_1, \mathbf{u}_n \rangle \\ \langle \mathbf{u}_2, \mathbf{u}_1 \rangle & \langle \mathbf{u}_2, \mathbf{u}_2 \rangle & \cdots & \langle \mathbf{u}_2, \mathbf{u}_n \rangle \\ \vdots & \vdots & \ddots & \vdots \\ \langle \mathbf{u}_n, \mathbf{u}_1 \rangle & \langle \mathbf{u}_n, \mathbf{u}_2 \rangle & \cdots & \langle \mathbf{u}_n, \mathbf{u}_n \rangle \end{bmatrix}$$

the **matrix of the inner product $\langle, \rangle$ relative to the basis $\mathcal{B}$** . Thus, using coordinate vectors

$$[\mathbf{u}]_{\mathcal{B}} = [x_1, x_2, \dots, x_n]^T, \quad [\mathbf{v}]_{\mathcal{B}} = [y_1, y_2, \dots, y_n]^T,$$

we have

$$\langle \mathbf{u}, \mathbf{v} \rangle = [\mathbf{u}]_{\mathcal{B}}^T A [\mathbf{v}]_{\mathcal{B}}.$$

4 Representation of inner product

Theorem 4.1. *Let V be an n -dimensional vector space with an inner product $\langle, \rangle$, and let A be the matrix of $\langle, \rangle$ relative to a basis $\mathcal{B}$. Then for any vectors $\mathbf{u}, \mathbf{v} \in V$,*

$$\langle \mathbf{u}, \mathbf{v} \rangle = \mathbf{x}^T A \mathbf{y}.$$

where $\mathbf{x}$ and $\mathbf{y}$ are the coordinate vectors of $\mathbf{u}$ and $\mathbf{v}$, respectively, i.e., $\mathbf{x} = [\mathbf{u}]_{\mathcal{B}}$ and $\mathbf{y} = [\mathbf{v}]_{\mathcal{B}}$.

Example 4.1. For the inner product of $\mathbb{R}^3$ defined by

$$\langle \mathbf{x}, \mathbf{y} \rangle = 2x_1y_1 - x_1y_2 - x_2y_1 + 5x_2y_2,$$

where $\mathbf{x} = \begin{bmatrix} x_1 \\ x_2 \end{bmatrix}, \mathbf{y} = \begin{bmatrix} y_1 \\ y_2 \end{bmatrix} \in \mathbb{R}^2$, its matrix relative to the standard basis $\mathcal{E} = \{\mathbf{e}_1, \mathbf{e}_2\}$ is

$$A = \begin{bmatrix} \langle \mathbf{e}_1, \mathbf{e}_1 \rangle & \langle \mathbf{e}_1, \mathbf{e}_2 \rangle \\ \langle \mathbf{e}_2, \mathbf{e}_1 \rangle & \langle \mathbf{e}_2, \mathbf{e}_2 \rangle \end{bmatrix} = \begin{bmatrix} 2 & -1 \\ -1 & 5 \end{bmatrix}.$$

The inner product can be written as

$$\langle \mathbf{x}, \mathbf{y} \rangle = \mathbf{x}^T A \mathbf{y} = [x_1, x_2] \begin{bmatrix} 2 & -1 \\ -1 & 5 \end{bmatrix} \begin{bmatrix} y_1 \\ y_2 \end{bmatrix}.$$

Let V be a real inner product space with basis $S = \{u_1, u_2, \dots, u_n\}$. The matrix

$$A = [a_{ij}], \quad \text{where} \quad a_{ij} = \langle u_i, u_j \rangle$$

is called the *matrix representation of the inner product on V relative to the basis S* .

Observe that A is symmetric, because the inner product is symmetric; that is, $\langle u_i, u_j \rangle = \langle u_j, u_i \rangle$. Also, A depends on both the inner product on V and the basis S for V . Moreover, if S is an orthogonal basis, then A is diagonal, and if S is an orthonormal basis, then A is the identity matrix.

EXAMPLE 7.14 The vectors $u_1 = (1, 1, 0)$, $u_2 = (1, 2, 3)$, $u_3 = (1, 3, 5)$ form a basis S for Euclidean space $\mathbf{R}^3$. Find the matrix A that represents the inner product in $\mathbf{R}^3$ relative to this basis S .

First compute each $\langle u_i, u_j \rangle$ to obtain

$$\begin{aligned} \langle u_1, u_1 \rangle &= 1 + 1 + 0 = 2, & \langle u_1, u_2 \rangle &= 1 + 2 + 0 = 3, & \langle u_1, u_3 \rangle &= 1 + 3 + 0 = 4 \\ \langle u_2, u_2 \rangle &= 1 + 4 + 9 = 14, & \langle u_2, u_3 \rangle &= 1 + 6 + 15 = 22, & \langle u_3, u_3 \rangle &= 1 + 9 + 25 = 35 \end{aligned}$$

Then $A = \begin{bmatrix} 2 & 3 & 4 \\ 3 & 14 & 22 \\ 4 & 22 & 35 \end{bmatrix}$. As expected, A is symmetric.

Orthogonal Matrices

- 7.79.** Find the number and exhibit all 2×2 orthogonal matrices of the form $\begin{bmatrix} \frac{1}{3} & x \\ y & z \end{bmatrix}$.
- 7.80.** Find a 3×3 orthogonal matrix P whose first two rows are multiples of $u = (1, 1, 1)$ and $v = (1, -2, 3)$, respectively.
- 7.81.** Find a symmetric orthogonal matrix P whose first row is $(\frac{1}{3}, \frac{2}{3}, \frac{2}{3})$. (Compare with Problem 7.32.)
- 7.82.** Real matrices A and B are said to be *orthogonally equivalent* if there exists an orthogonal matrix P such that $B = P^T A P$. Show that this relation is an equivalence relation.

Positive Definite Matrices and Inner Products

- 7.83.** Find the matrix A that represents the usual inner product on $\mathbf{R}^2$ relative to each of the following bases:

(a) $\{v_1 = (1, 4), v_2 = (2, -3)\}$, (b) $\{w_1 = (1, -3), w_2 = (6, 2)\}$.

- 7.84.** Consider the following inner product on $\mathbf{R}^2$:

$$f(u, v) = x_1 y_1 - 2x_1 y_2 - 2x_2 y_1 + 5x_2 y_2, \quad \text{where} \quad u = (x_1, x_2) \quad v = (y_1, y_2)$$

Find the matrix B that represents this inner product on $\mathbf{R}^2$ relative to each basis in Problem 7.83.

- 7.85.** Find the matrix C that represents the usual basis on $\mathbf{R}^3$ relative to the basis S of $\mathbf{R}^3$ consisting of the vectors $u_1 = (1, 1, 1)$, $u_2 = (1, 2, 1)$, $u_3 = (1, -1, 3)$.

- 7.86.** Let $V = \mathbf{P}_2(t)$ with inner product $\langle f, g \rangle = \int_0^1 f(t)g(t) dt$.

- (a) Find $\langle f, g \rangle$, where $f(t) = t + 2$ and $g(t) = t^2 - 3t + 4$.
- (b) Find the matrix A of the inner product with respect to the basis $\{1, t, t^2\}$ of V .
- (c) Verify Theorem 7.16 that $\langle f, g \rangle = [f]^T A [g]$ with respect to the basis $\{1, t, t^2\}$.

$$R_2 \rightarrow R_2 - R_1, R_3 \rightarrow R_3 - 3R_1, R_4 \rightarrow R_4 - 2R_1$$

$$\sim \begin{bmatrix} 1 & 2 & 2 \\ 0 & -1 & 1 \\ 0 & -3 & 0 \\ 0 & -2 & 0 \end{bmatrix}, R_3 \rightarrow R_3 - 3R_2, R_4 \rightarrow R_4 - 2R_2 \sim \begin{bmatrix} 1 & 2 & 2 \\ 0 & -1 & 1 \\ 0 & 0 & -3 \\ 0 & 0 & -2 \end{bmatrix}$$

$$R_4 \rightarrow R_4 - \frac{2}{3}R_3$$

$$\sim \begin{bmatrix} 1 & 2 & 2 \\ 0 & -1 & 1 \\ 0 & 0 & -3 \\ 0 & 0 & 0 \end{bmatrix} \Rightarrow \rho(A) = 3 \text{ and } n = 3$$

$\therefore$ The system has a trivial solution

$$\text{Hence, } a_1 = a_2 = a_3 = 0$$

i.e., they are linearly independent.

EXERCISE 3.5

- Examine the following vectors for linear dependence and find the relation if it exists.
 $X_1 = (1, 2, 4)$, $X_2 = (2, -1, 3)$, $X_3 = (0, 1, 2)$, $X_4 = (-3, 7, 2)$. (U.P.T.U., 2002)
 [Ans. Linearly dependent, $9X_1 - 12X_2 + 5X_3 - 5X_4 = 0$]
- Show the vectors $X_1 = [1, 2, 1]$, $X_2 = [2, 1, 4]$, $X_3 = [4, 5, 6]$ and $X_4 = [1, 8, -3]$ are linearly independent?
- Show that the vectors $[1, 2, 3]$, $[3, -2, 1]$, $[1, -6, -5]$ are linearly dependent.
- If the vectors $(0, 1, a)$, $(1, a, 1)$, $(a, 1, 0)$ are linearly dependent, then find the value of a .
 [Ans. $a = 0, \sqrt{2}, -\sqrt{2}$]
- Examine for linear dependence $[1, 0, 2, 1]$, $[3, 1, 2, 1]$, $[4, 6, 2, -4]$, $[-6, 0, -3, -4]$ and find the relation between them, if possible.
 [Ans. Linear dependent and the relation is $2X_1 - 6X_2 + X_3 - 2X_4 = 0$]
- Show that the vectors $X_1 = [2, i, -i]$, $X_2 = [2i, -1, 1]$, $X_3 = [1, 2, 3]$ are linearly dependent.
- If $X_1 = [1, 1, 2]$, $X_2 = [2, -1, -6]$, $X_3 = [13, 4, -4]$ prove that $7X_1 + 3X_2 - X_3 = 0$.
- Show that the vectors $[3, 1, -4]$, $[2, 2, -3]$ form a linearly independent set but $[3, 1, -4]$, $[2, 2, -3]$ and $[0, -4, 1]$ are linearly dependent.

3.17 EIGEN VALUES AND EIGEN VECTORS

Introduction: At the start of 20th century, Hilbert studied the eigen values. He was the first to use the German word *eigen* to denote eigen value and eigen vectors in 1904. The word *eigen* mean—**own characteristic or individual**.

More formally in a vector space, a vector function A (matrix) defined if each vector X of vector space, there corresponds a unique vector $Y = AX$. So here we consider a linear transform $Y = AX$

transforms X into a scalar multiple of itself say λ so $AX = \lambda X$ which is called **Eigen value equation**.

In this section, we study the problem

$$AX = \lambda X \quad \dots(i)$$

where A is a $n \times n$ matrix, X is an unknown $n \times 1$ vector and λ is an unknown scalar. From equation (i) it can be understood that AX is a scalar multiple of X say λX . Geometrically each vector on the line through the origin determined by X gets mapped back onto the same line under multiplication by A .

Geometrical representation: The eigen value equation means that under the transformation, a eigen vector experience only changes in magnitude and sign. The direction of AX is the same as that of X .

Here A acts to stretch the vector X , not change its direction. So X is an given vector of A .

The eigen value determines the amount, the eigen vector is scaled under the linear transformation. For example, the eigen value $\lambda = 2$ means that the eigen vector is doubled in length and point in the same direction. The eigen value $\lambda = 1$, means that the eigen vector is unchanged, while an eigen value $\lambda = -1$ means that the eigen vector is reversed in direction.

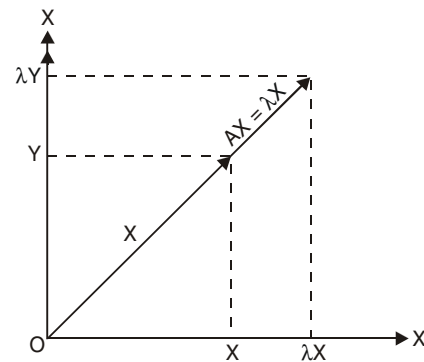


Fig. 3.1

Thus, the eigen value λ is simply the amount of “stretches” or “shrinks” to which a vector is subjected when transformed by A .

3.17.1 Characteristic Equation

(U.P.T.U., 2007)

If we re-express (i) as $AX = \lambda IX$ (where I is an identity matrix).

or

$$AX - \lambda IX = O$$

$\Rightarrow$

$$(A - \lambda I)X = O \quad \dots(ii)$$

Which is homogeneous system of n equations in the n variables $x_1, x_2, \dots, x_n$. The system (ii) must have non-trivial solutions otherwise $X = 0$ (which is impossible).

$\therefore$ For non-trivial solution the coefficient matrix $(A - \lambda I)$ will be singular

$$\Rightarrow |A - \lambda I| = 0 \quad \dots(iii) \text{ |singular matrix } |A| = 0$$

Expansion of the determinant gives an algebraic equation in λ , known as the “characteristic equation” of A . The determinant $|A - \lambda I|$ is called characteristic polynomial of A .

3.17.2 Characteristic Roots or Eigen Values

[U.P.T.U. (C.O.), 2003, 2007]

The roots of characteristic equation are called characteristic roots or eigen values.

3.17.3 Eigen Vectors

[U.P.T.U. (C.O.), 2003, 2007]

The corresponding non-zero vector X is called characteristic eigen vector.

- Notes:**
1. If there is one linearly independent solution and two eigen values are same then there will be same eigen vectors of both eigen values.
 2. If there is two linearly independent solution and two eigen values are same then there will be different eigen vectors of each eigen value.

3.17.4 Properties of Eigen Values and Eigen Vectors

1. If A is real, its eigen values are real or complex conjugate in pairs.
2. Determinant of A = product of eigen values of A .
3. A and A^T has same eigen values.
4. A^{-1} exists iff 0 is not an eigen value of A , eigen values of A^{-1} are $\frac{1}{\lambda_1}, \frac{1}{\lambda_2}, \dots, \frac{1}{\lambda_n}$.
(U.P.T.U., 2008)
5. Eigen vector cannot correspond to two distinct characteristic values.
6. **Eigen values of diagonal**, upper triangular or lower triangular matrices are the principal diagonal elements.
7. KA (scalar multiples) has eigen values $K\lambda_i$.
8. A^m has eigen values λ^m .
9. Two vectors X and Y are said to be orthogonal if $X^T Y = Y^T X = 0$. (U.P.T.U., 2008)

Theorem 1. The latent roots of a Hermitian matrix are all real [U.P.T.U. (C.O.), 2003]

Proof. We have $AX = \lambda X$... (i)

To prove that λ is a real number, we have to prove $\lambda = \bar{\lambda}$

From (i)

$$\begin{aligned} X^*(AX) &= X^*(\lambda X) \Rightarrow X^*AX = \lambda X^*X \\ \Rightarrow (X^*AX)^* &= (\lambda X^*X)^* \Rightarrow X^*A^*X = \bar{\lambda} X^*X, (X^{**} = X) \\ \Rightarrow X^*AX &= \bar{\lambda} X^*X (A^* = A) \\ \Rightarrow X^*\lambda X &= \bar{\lambda} X^*X \Rightarrow \lambda X^*X = \bar{\lambda} X^*X \\ \Rightarrow (\lambda - \bar{\lambda}) X^*X &= 0 \Rightarrow \lambda = \bar{\lambda}. \text{ Proved.} \end{aligned}$$

Theorem 2. The latent roots of a skew-Hermitian matrix are either zero or purely imaginary. [U.P.T.U. (C.O.), 2003]

Proof. Let A be a skew-Hermitian matrix so that $A^* = -A$.

Now we are to prove that $\lambda = 0$ or purely imaginary number

Here $(iA)^* = \bar{i} A^* = -iA^* = -i(-A) = iA$ (As $A^* = -A$)

Hence, iA is a Hermitian matrix.

Let λ be an eigen value relative to the eigen vector X of A , then

$$\begin{aligned} AX &= \lambda X \quad \dots (i) \\ \Rightarrow iAX &= i\lambda X \Rightarrow (iA)X = (i\lambda)X \\ \Rightarrow i\lambda &\text{ is an eigen root relative to the vector } X \text{ of Hermitian matrix } iA. \\ \Rightarrow i\lambda &\text{ is a real number, for eigen values of a Hermitian matrix are all real.} \\ \Rightarrow \lambda &= 0 \text{ or purely imaginary number. Proved.} \end{aligned}$$

Theorem 3. The characteristic roots of a unitary matrix are of unit modulus.

[U.P.T.U. Special Exam., 2001]

Proof. Let A be unitary matrix so that $A^*A = I$.

We have $AX = \lambda X$... (i)

To prove $|\lambda| = 1$

From (i), we have

$$(AX)^* = (\lambda X)^* \Rightarrow X^* A^* = \bar{\lambda} X^* \quad \dots(ii)$$

Pre-multiplying (i) by (ii),

$$(X^* A^*) (AX) = (\bar{\lambda} X^*) (\lambda X)$$

$$\Rightarrow X^* (A^* A) X = \bar{\lambda} \lambda X^* X$$

$$\Rightarrow X^* I X = |\lambda|^2 X^* X \Rightarrow (1 - |\lambda|^2) X^* X = 0$$

$$\Rightarrow 1 - |\lambda|^2 = 0 \Rightarrow |\lambda|^2 = 1 \text{ or } |\lambda| = 1. \text{ **Proved.**}$$

Theorem 4. Prove that the product of all eigen values of A is equal to the determinant $|A|$.
(U.P.T.U., 2004)

Proof. We have $AX = \lambda X \Rightarrow (A - \lambda I) X = 0$

For non-trivial solution $|A - \lambda I| = 0$

$$\Rightarrow \begin{vmatrix} a_{11} - \lambda & a_{12} & \dots & a_{1n} \\ \vdots & & & \\ a_{n1} & a_{n2} & \dots & a_{nn} - \lambda \end{vmatrix} = 0$$

$$\Rightarrow (-1)^n \{\lambda^n + b_1 \lambda^{n-1} + b_2 \lambda^{n-2} + \dots + b_n\} = 0$$

$$\Rightarrow (\lambda - \lambda_1) (\lambda - \lambda_2) \dots (\lambda - \lambda_n) = 0$$

putting $\lambda = 0$, we get

$$|A| = \lambda_1 \lambda_2 \lambda_3 \dots \lambda_n. \text{ **Proved.**}$$

Theorem 5. The latent roots of real symmetric matrix are all real.

Proof. Let A be a real symmetric matrix so that

$$\bar{A} = A, A' = A.$$

The $A^* = (\bar{A})' = A' = A$ or $A^* = A$, meaning thereby, A is a Hermitian matrix. Hence, the latent roots of A are all real, by Theorem 1.

Theorem 6. The characteristic roots of an idempotent matrix are either zero or unity.

Proof. Let A be idempotent matrix so that $A^2 = A$.

$$\text{Let } AX = \lambda X \quad \dots(i)$$

premultiplying by A on equation (i), we get

$$A(AX) = A(\lambda X) = \lambda(AX)$$

$$\Rightarrow (AA)X = \lambda(\lambda X) \Rightarrow A^2 X = \lambda^2 X \text{ or } AX = \lambda^2 X \quad (\text{As } A^2 = A)$$

$$\Rightarrow \lambda X = \lambda^2 X$$

$$\Rightarrow (\lambda^2 - \lambda)X = 0 \Rightarrow \lambda^2 - \lambda = 0 \quad (\text{As } X \neq 0)$$

$$\Rightarrow \lambda = 0, 1. \text{ **Proved.**}$$

Example 1. Show that 0 is a characteristic root of a matrix if the matrix is singular.

(U.P.T.U., 2008)

Sol. The characteristic equation is

$$|A - \lambda I| = 0$$

$$\text{or } |A| - \lambda |I| = 0$$

$$\text{As } |A| = 0 \text{ (singular matrix)}$$

$$\therefore 0 = \lambda |I| = 0 \Rightarrow \lambda = 0$$

Conversely, if $\lambda = 0$ then $|A| = 0$. **Proved.**

Example 2. Find the characteristic equation of the matrix $\begin{bmatrix} 1 & 2 & 2 \\ 0 & 2 & 1 \\ -1 & 2 & 2 \end{bmatrix}$. Also find the eigen values and eigen vectors of this matrix. (U.P.T.U., 2006)

Sol. Let
$$A = \begin{bmatrix} 1 & 2 & 2 \\ 0 & 2 & 1 \\ -1 & 2 & 2 \end{bmatrix}$$

Now,
$$A - \lambda I = \begin{bmatrix} 1 & 2 & 2 \\ 0 & 2 & 1 \\ -1 & 2 & 2 \end{bmatrix} - \lambda \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} = \begin{bmatrix} 1-\lambda & 2 & 2 \\ 0 & 2-\lambda & 1 \\ -1 & 2 & 2-\lambda \end{bmatrix}$$

$\therefore$ Characteristic equation is $|A - \lambda I| = 0$

$$\Rightarrow \begin{vmatrix} 1-\lambda & 2 & 2 \\ 0 & 2-\lambda & 1 \\ -1 & 2 & 2-\lambda \end{vmatrix} = 0 \Rightarrow (1-\lambda) \{(2-\lambda)^2 - 2\} - 2\{0+1\} + 2\{0+(2-\lambda)\} = 0$$

$$\Rightarrow (1-\lambda)(\lambda^2 - 4\lambda + 2) - 2 + 4 - 2\lambda = 0$$

$$\Rightarrow \lambda^2 - 4\lambda + 2 - \lambda^3 + 4\lambda^2 - 2\lambda + 2 - 2\lambda = 0$$

$$\Rightarrow \lambda^3 - 5\lambda^2 + 8\lambda - 4 = 0$$

$$\Rightarrow \lambda^2(\lambda - 1) - 4\lambda(\lambda - 1) + 4(\lambda - 1) = 0$$

$$\Rightarrow (\lambda - 1)(\lambda^2 - 4\lambda + 4) = 0$$

Hence, $\lambda = 1, 2, 2$. These are eigen values of A .

Now, we consider the relation

$$AX = \lambda X \Rightarrow AX = \lambda IX$$

$$\Rightarrow (A - \lambda I)X = 0 \quad \dots(i)$$

Taking $\lambda = 1$, from (1), we get

$$\begin{bmatrix} 1-1 & 2 & 2 \\ 0 & 2-1 & 1 \\ -1 & 2 & 2-1 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = 0$$

$$\Rightarrow \begin{bmatrix} 0 & 2 & 2 \\ 0 & 1 & 1 \\ -1 & 2 & 1 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = 0$$

$$R_1 \leftrightarrow R_3$$

$$\sim \begin{bmatrix} -1 & 2 & 1 \\ 0 & 1 & 1 \\ 0 & 2 & 2 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = 0$$

$$R_3 \rightarrow R_3 - 2R_2, R_1 \rightarrow (-1) R_1$$

$$\sim \begin{bmatrix} 1 & -2 & -1 \\ 0 & 1 & 1 \\ 0 & 0 & 0 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = 0$$

$$\Rightarrow \begin{aligned} x_1 - 2x_2 - x_3 &= 0 \\ x_2 + x_3 &= 0 \end{aligned}$$

$$\text{Let } x_3 = k \text{ then } x_2 = -k$$

$$\text{and } x_1 + 2k - k = 0 \Rightarrow x_1 = -k$$

$$\therefore X_1 = \begin{bmatrix} -k \\ -k \\ k \end{bmatrix} = -k \begin{bmatrix} 1 \\ 1 \\ -1 \end{bmatrix} \text{ or } \begin{bmatrix} 1 \\ 1 \\ -1 \end{bmatrix}$$

Taking $\lambda = 2$, from (1), we get

$$\begin{bmatrix} -1 & 2 & 2 \\ 0 & 0 & 1 \\ -1 & 2 & 0 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = 0$$

$$R_2 \leftrightarrow R_3$$

$$\sim \begin{bmatrix} -1 & 2 & 2 \\ -1 & 2 & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = 0$$

$$R_2 \rightarrow R_2 - R_1, R_1 \rightarrow -R_1$$

$$\sim \begin{bmatrix} 1 & -2 & -2 \\ 0 & 0 & -2 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = 0$$

$$R_2 \rightarrow -\frac{1}{2}R_2 \text{ and then } R_3 \rightarrow R_3 - R_2, \text{ we get}$$

$$\sim \begin{bmatrix} 1 & -2 & -2 \\ 0 & 0 & 1 \\ 0 & 0 & 0 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = 0$$

Here the solution is one variable linearly independent.

$$\Rightarrow \begin{aligned} x_1 - 2x_2 - x_3 &= 0 \\ x_3 &= 0 \end{aligned}$$

$$\text{Let } x_2 = k \text{ then } x_1 - 2k - 0 = 0$$

$$\Rightarrow x_1 = 2k$$

$$\text{Hence, } X_2 = \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 2k \\ k \\ 0 \end{bmatrix} = k \begin{bmatrix} 2 \\ 1 \\ 0 \end{bmatrix} \text{ or } \begin{bmatrix} 2 \\ 1 \\ 0 \end{bmatrix}$$

Thus only one eigen vector X_2 corresponds to the repeated eigen value $\lambda = 2$.

Hence the eigen values are 1, 2, 2

eigen vectors
$$X_1 = -k \begin{bmatrix} 1 \\ 1 \\ -1 \end{bmatrix}, \quad X_2 = X_3 = k \begin{bmatrix} 2 \\ 1 \\ 0 \end{bmatrix}$$

or
$$X_1 = \begin{bmatrix} 1 \\ 1 \\ -1 \end{bmatrix}, \quad X_2 = X_3 = \begin{bmatrix} 2 \\ 1 \\ 0 \end{bmatrix}$$

Example 3. Find the eigen values and eigen vectors of the matrix

$$A = \begin{bmatrix} 2 & 1 & 1 \\ 1 & 2 & 1 \\ 0 & 0 & 1 \end{bmatrix}$$

Sol. The characteristic equation is $|A - \lambda I| = 0$

$$\Rightarrow \begin{vmatrix} 2-\lambda & 1 & 1 \\ 1 & 2-\lambda & 1 \\ 0 & 0 & 1-\lambda \end{vmatrix} = (1-\lambda)(\lambda-1)(\lambda-3) = 0$$

Thus $\lambda = 1, 1, 3$ are the eigen values of A .

Now, we consider the relation $(A - \lambda I)X = 0$

For $\lambda = 3$,

$$\begin{bmatrix} -1 & 1 & 1 \\ 1 & -1 & 1 \\ 0 & 0 & -2 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = 0 \quad \left| \text{let } X = \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} \right.$$

Applying $R_2 \rightarrow R_2 + R_1$, then $R_1 \rightarrow -R_1$ on coefficient matrix, we get

$$\sim \begin{bmatrix} 1 & -1 & -1 \\ 0 & 0 & 2 \\ 0 & 0 & -2 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = 0$$

Again $R_3 \rightarrow R_3 + R_2$, $R_2 \rightarrow \frac{1}{2}R_2$

$$\sim \begin{bmatrix} 1 & -1 & -1 \\ 0 & 0 & 1 \\ 0 & 0 & 0 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = 0$$

$$\Rightarrow x_1 - x_2 - x_3 = 0$$

$$x_3 = 0 \quad \left| \begin{array}{l} r=2, n=3 \\ n-r=1 \end{array} \right.$$

Suppose $x_2 = k$, then $x_1 = k$

Here,

$$X_1 = \begin{bmatrix} k \\ k \\ 0 \end{bmatrix} = k \begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix} \text{ or } \begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix}$$

For $\lambda = 1$

$$\begin{bmatrix} 1 & 1 & 1 \\ 1 & 1 & 1 \\ 0 & 0 & 0 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = 0$$

$R_2 \rightarrow R_2 - R_1$

$$\sim \begin{bmatrix} 1 & 1 & 1 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = 0$$

Here $r = 1, n = 3 \Rightarrow n - r = 3 - 1 = 2$.

$\therefore$ There is two variables linearly independent solution

Let $x_2 = k_1, x_3 = k_2$ and $x_1 + x_2 + x_3 = 0$
 $\Rightarrow x_1 + k_1 + k_2 = 0 \Rightarrow x_1 = -(k_1 + k_2)$

$$\Rightarrow X_2 = \begin{bmatrix} -(k_1 + k_2) \\ k_1 \\ k_2 \end{bmatrix}.$$

Since the vectors are linearly independent so, we choose k_1 and k_2 as follows:

(a) If we suppose $x_2 = k_1 = 0, x_3 = k_2$ (any arbitrary)

Then

$$X_2 = \begin{bmatrix} -k_2 \\ 0 \\ k_2 \end{bmatrix} = -k_2 \begin{bmatrix} 1 \\ 0 \\ -1 \end{bmatrix} \text{ or } \begin{bmatrix} 1 \\ 0 \\ -1 \end{bmatrix}$$

(b) If we suppose $x_2 = k_1$ (any arbitrary) and $x_3 = k_2 = 0$

Then

$$X_3 = \begin{bmatrix} -k_1 \\ k_1 \\ 0 \end{bmatrix} = -k_1 \begin{bmatrix} 1 \\ -1 \\ 0 \end{bmatrix} \text{ or } \begin{bmatrix} 1 \\ -1 \\ 0 \end{bmatrix}$$

Hence the eigen values are $\lambda = 1, 2, 2$

Eigen vectors are
$$X_1 = k \begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix}; X_2 = -k_2 \begin{bmatrix} 1 \\ 0 \\ -1 \end{bmatrix}; X_3 = -k_1 \begin{bmatrix} 1 \\ -1 \\ 0 \end{bmatrix}$$

or

$$X_1 = \begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix}; X_2 = \begin{bmatrix} 1 \\ 0 \\ -1 \end{bmatrix}; X_3 = \begin{bmatrix} 1 \\ -1 \\ 0 \end{bmatrix}.$$

Example 4. Find the eigen values and eigen-vectors of matrix

$$A = \begin{bmatrix} 3 & 1 & 4 \\ 0 & 2 & 6 \\ 0 & 0 & 5 \end{bmatrix}. \quad [U.P.T.U., 2004 (C.O.), 2002]$$

Sol. The characteristic equation is $|A - \lambda I| = 0$

$$\Rightarrow \begin{vmatrix} 3-\lambda & 1 & 4 \\ 0 & 2-\lambda & 6 \\ 0 & 0 & 5-\lambda \end{vmatrix} = (3-\lambda)\{(2-\lambda)(5-\lambda)\} - 0 + 0 = 0$$

$$\therefore \lambda = 2, 3, 5$$

Now, we consider the relation $(A - \lambda I) X = 0$

...(i)

For $\lambda = 2$,

$$\begin{bmatrix} 3-2 & 1 & 4 \\ 0 & 2-2 & 6 \\ 0 & 0 & 5-2 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = 0 \quad \left| X = \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} \right.$$

$$\Rightarrow \begin{bmatrix} 1 & 1 & 4 \\ 0 & 0 & 6 \\ 0 & 0 & 3 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = 0$$

$R_3 \rightarrow R_3 - \frac{1}{2}R_2$ on coefficient matrix

$$\sim \begin{bmatrix} 1 & 1 & 4 \\ 0 & 0 & 6 \\ 0 & 0 & 0 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = 0$$

$$\Rightarrow \begin{array}{ll} x_1 + x_2 + 4x_3 = 0 & \text{and } x_3 = 0 \\ \text{Let } x_2 = k & \text{then } x_1 + k + 0 = 0 \Rightarrow x_1 = -k \end{array}$$

$$\therefore X_1 = \begin{bmatrix} -k \\ k \\ 0 \end{bmatrix} = -k \begin{bmatrix} 1 \\ -1 \\ 0 \end{bmatrix} \text{ or } \begin{bmatrix} 1 \\ -1 \\ 0 \end{bmatrix}$$

For $\lambda = 3$, from (i), we get

$$\begin{bmatrix} 0 & 1 & 4 \\ 0 & -1 & 6 \\ 0 & 0 & 2 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = 0$$

$R_2 \rightarrow R_2 + R_1$ on coefficient matrix

$$\sim \begin{bmatrix} 0 & 1 & 4 \\ 0 & 0 & 10 \\ 0 & 0 & 2 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = 0$$

$R_3 \rightarrow R_3 - \frac{1}{5}R_2$

$$\sim \begin{bmatrix} 0 & 1 & 4 \\ 0 & 0 & 10 \\ 0 & 0 & 0 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = 0$$

$$\Rightarrow x_2 + 4x_3 = 0 \text{ and } 10x_3 = 0 \Rightarrow x_3 = 0$$

$$\Rightarrow x_2 + 0 = 0 \Rightarrow x_2 = 0, \text{ let } x_1 = K$$

$$\therefore X_2 = \begin{bmatrix} k \\ 0 \\ 0 \end{bmatrix} = k \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix} \text{ or } \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}$$

Again, for $\lambda = 5$, we get

$$\begin{bmatrix} -2 & 1 & 4 \\ 0 & -3 & 6 \\ 0 & 0 & 0 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$

$$\Rightarrow 2x_1 - x_2 - 4x_3 = 0$$

$$3x_2 - 6x_3 = 0$$

$$\text{Let } x_3 = k, \text{ then } 3x_2 - 6k = 0$$

$$\Rightarrow x_2 = 2k \text{ and } 2x_1 - 2k - 4k = 0 \Rightarrow x_1 = 3k$$

$$\therefore X_3 = \begin{bmatrix} 3k \\ 2k \\ k \end{bmatrix} = k \begin{bmatrix} 3 \\ 2 \\ 1 \end{bmatrix} \text{ or } \begin{bmatrix} 3 \\ 2 \\ 1 \end{bmatrix}$$

Hence the eigen values are $\lambda = 2, 3, 5$

$$\text{And eigen vectors are } X_1 = \begin{bmatrix} 1 \\ -1 \\ 0 \end{bmatrix}, X_2 = \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}, X_3 = \begin{bmatrix} 3 \\ 2 \\ 1 \end{bmatrix}.$$

Example 5. Determine the latent roots and the corresponding vector of the matrix,

$$A = \begin{bmatrix} 6 & -2 & 2 \\ -2 & 3 & -1 \\ 2 & -1 & 3 \end{bmatrix}.$$

Sol. The characteristic equation is $|A - \lambda I| = 0$

$$\Rightarrow \begin{vmatrix} 6-\lambda & -2 & 2 \\ -2 & 3-\lambda & -1 \\ 2 & -1 & 3-\lambda \end{vmatrix} = 0$$

$R_2 \rightarrow R_2 + R_3$, we get

$$\begin{vmatrix} 6-\lambda & -2 & 2 \\ 0 & 2-\lambda & 2-\lambda \\ 2 & -1 & 3-\lambda \end{vmatrix} = (2-\lambda) \begin{vmatrix} 6-\lambda & -2 & 2 \\ 0 & 1 & 1 \\ 2 & -1 & 3-\lambda \end{vmatrix} = 0$$

Now, $C_3 \rightarrow C_3 + C_2$ gives

$$(2-\lambda) \begin{vmatrix} 6-\lambda & -2 & 0 \\ 0 & 1 & 2 \\ 2 & -1 & 2-\lambda \end{vmatrix} = (2-\lambda) \{(6-\lambda)(2-\lambda+2)-8\} = 0$$

$$\therefore (2-\lambda)(\lambda^2 - 10\lambda + 16) = 0 \Rightarrow (2-\lambda)(2-\lambda)(\lambda-8) = 0$$

$$\Rightarrow \lambda = 2, 2, 8$$

Now, consider the relation

$$(A - \lambda I)X = 0$$

...(i)

For $\lambda = 8$, the equation (i), gives

$$\begin{bmatrix} 6-8 & -2 & 2 \\ -2 & 3-8 & -1 \\ 2 & -1 & 3-8 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$

$$\Rightarrow \begin{bmatrix} -2 & -2 & 2 \\ -2 & -5 & -1 \\ 2 & -1 & -5 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$

$R_2 \rightarrow R_2 - R_1$, $R_3 \rightarrow R_3 + R_1$ on coefficient matrix

$$\begin{bmatrix} -2 & -2 & 2 \\ 0 & -3 & -3 \\ 0 & -3 & -3 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$

$$R_3 \rightarrow R_3 - R_2 \text{ and } R_1 \rightarrow \frac{-1}{2}R_1, R_2 \rightarrow \frac{-1}{3}R_2$$

$$\begin{bmatrix} 1 & 1 & -1 \\ 0 & 1 & 1 \\ 0 & 0 & 0 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$

$$\Rightarrow \begin{aligned} x_1 + x_2 - x_3 &= 0 \\ x_2 + x_3 &= 0 \end{aligned}$$

Let $x_3 = k$, then $x_2 = -k$ and $x_1 = 2k$

$$\therefore X_1 = \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 2k \\ -k \\ k \end{bmatrix} = k \begin{bmatrix} 2 \\ -1 \\ 1 \end{bmatrix} \text{ or } \begin{bmatrix} 2 \\ -1 \\ 1 \end{bmatrix}$$

For $\lambda = 2$, the equation (i) gives

$$\begin{bmatrix} 6-2 & -2 & 2 \\ -2 & 3-2 & -1 \\ 2 & -1 & 3-2 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$

$$\Rightarrow \begin{bmatrix} 4 & -2 & 2 \\ -2 & 1 & -1 \\ 2 & -1 & 1 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$

$R_2 \rightarrow R_2 + \frac{1}{2}R_1, R_3 \rightarrow R_3 - \frac{1}{2}R_1$, we get

$$\begin{bmatrix} 4 & -2 & 2 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$

Here $r = 1, n = 3 \Rightarrow n - r = 3 - 1 = 2$ (Two linearly independent)

Let $x_2 = k_1$ and $x_3 = k_2$

and we have $4x_1 - 2x_2 + 2x_3 = 0$
 $\Rightarrow 4x_1 - 2k_1 + 2k_2 = 0$

$$\Rightarrow x_1 = \frac{1}{2}(k_1 - k_2)$$

$$\therefore X_2 = \begin{bmatrix} k_1 \\ k_2 \\ \frac{1}{2}(k_1 - k_2) \end{bmatrix} = \frac{1}{2} \begin{bmatrix} 2k_1 \\ 2k_2 \\ (k_1 - k_2) \end{bmatrix}$$

Since the vectors are linearly independent so, we choose, k_1 and k_2 as follows:

(i) Let $x_2 = k_1 = 0$ and $x_3 = k_2$ (arbitrary)

then
$$X_2 = \frac{1}{2} \begin{bmatrix} 0 \\ 2k_2 \\ -k_2 \end{bmatrix} = \frac{k_2}{2} \begin{bmatrix} 0 \\ 2 \\ -1 \end{bmatrix} \text{ or } \begin{bmatrix} 0 \\ 2 \\ -1 \end{bmatrix}$$

(ii) Let $x_3 = k_2 = 0$ and $x_2 = k_1$ (arbitrary)

then
$$X_3 = \frac{1}{2} \begin{bmatrix} 2k_1 \\ 0 \\ k_1 \end{bmatrix} = \frac{k_1}{2} \begin{bmatrix} 2 \\ 0 \\ 1 \end{bmatrix} \text{ or } \begin{bmatrix} 2 \\ 0 \\ 1 \end{bmatrix}$$

Hence $\lambda = 8, 2, 2$

$$X_1 = k \begin{bmatrix} 2 \\ -1 \\ 1 \end{bmatrix} \text{ or } \begin{bmatrix} 2 \\ -1 \\ 1 \end{bmatrix}, X_2 = \frac{k_2}{2} \begin{bmatrix} 0 \\ 2 \\ -1 \end{bmatrix} \text{ or } \begin{bmatrix} 0 \\ 2 \\ -1 \end{bmatrix}, X_3 = \frac{k_1}{2} \begin{bmatrix} 2 \\ 0 \\ 1 \end{bmatrix} \text{ or } \begin{bmatrix} 2 \\ 0 \\ 1 \end{bmatrix}.$$

Example 6. Find the eigen values of matrix A .

$$A = \frac{1}{3} \begin{bmatrix} 1 & 2 & 2 \\ 2 & 1 & -2 \\ 2 & -2 & 1 \end{bmatrix}.$$

Sol. The characteristic equation of B is, where

$$B = \begin{bmatrix} 1 & 2 & 2 \\ 2 & 1 & -2 \\ 2 & -2 & 1 \end{bmatrix}$$

$$\begin{vmatrix} 1-\lambda & 2 & 2 \\ 2 & 1-\lambda & -2 \\ 2 & -2 & 1-\lambda \end{vmatrix} = \lambda^3 - 3\lambda^2 - 9\lambda + 27 = 0$$

or

$$(\lambda - 3)^2 (\lambda + 3) = 0 \Rightarrow \lambda = 3, 3, -3$$

So the eigen values of $A = \frac{1}{3}B = 1, 1, -1$.

Example 7. Show that the matrix

$$A = \begin{bmatrix} 3 & 10 & 5 \\ -2 & -3 & -4 \\ 3 & 5 & 7 \end{bmatrix}$$

has less than three independent eigen vectors. Is it possible to obtain a similarity transformation that will diagonalise this matrix. (U.P.T.U., 2001)

Sol. The characteristic equation is

$$|A - \lambda I| = 0$$

$$\text{or} \quad \begin{vmatrix} 3-\lambda & 10 & 5 \\ -2 & -3-\lambda & -4 \\ 3 & 5 & 7-\lambda \end{vmatrix} = 0$$

or

$$\lambda^3 - 7\lambda^2 + 16\lambda - 12 = 0$$

$$\Rightarrow (\lambda - 2)^2 (\lambda - 3) = 0 \Rightarrow \lambda = 2, 2, 3$$

Now, consider the relation

$$(A - \lambda I)X = 0 \quad \dots(i)$$

For $\lambda = 3$, we have

$$\begin{bmatrix} 3-3 & 10 & 5 \\ -2 & -3-3 & -4 \\ 3 & 5 & 7-3 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = 0$$

$$\Rightarrow \begin{bmatrix} 0 & 10 & 5 \\ -2 & -6 & -4 \\ 3 & 5 & 4 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = 0$$

$R_1 \rightarrow \frac{1}{5}R_1, R_2 \rightarrow -\frac{1}{2}R_2$, we have

$$\begin{bmatrix} 0 & 2 & 1 \\ 1 & 3 & 2 \\ 3 & 5 & 4 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = 0$$

$R_1 \leftrightarrow R_2$

$$\begin{bmatrix} 1 & 3 & 2 \\ 0 & 2 & 1 \\ 3 & 5 & 4 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = 0$$

$R_3 \rightarrow R_3 - 3R_1$ on coefficient matrix

$$\begin{bmatrix} 1 & 3 & 2 \\ 0 & 2 & 1 \\ 0 & -4 & -2 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = 0$$

$R_3 \rightarrow R_3 + 2R_2$

$$\begin{bmatrix} 1 & 3 & 2 \\ 0 & 2 & 1 \\ 0 & 0 & 0 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$

$$\Rightarrow \begin{aligned} x_1 + 3x_2 + 2x_3 &= 0 \\ 0x_1 + 2x_2 + x_3 &= 0 \end{aligned}$$

Solving these equations by cross-multiplication

$$\frac{x_1}{3-4} = -\frac{x_2}{1-0} = \frac{x_3}{2-0}$$

$$\Rightarrow \frac{x_1}{-1} = \frac{x_2}{-1} = \frac{x_3}{2}$$

$$\Rightarrow \frac{x_1}{1} = \frac{x_2}{1} = \frac{x_3}{-2}$$

$\therefore$ The proportional values are

$$x_1 = 1, x_2 = 1, x_3 = -2$$

$$X_1 = \begin{bmatrix} 1 \\ 1 \\ -2 \end{bmatrix}$$

For $\lambda = 2$, we have, from equation (i)

$$\begin{bmatrix} 3-2 & 10 & 5 \\ -2 & -3-2 & -4 \\ 3 & 5 & 7-2 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = 0$$

$$\begin{bmatrix} 1 & 10 & 5 \\ -2 & -5 & -4 \\ 3 & 5 & 5 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = 0$$

Applying $R_2 \rightarrow R_2 + 2R_1$, $R_3 \rightarrow R_3 - 3R_1$ on coefficient matrix

$$\begin{bmatrix} 1 & 10 & 5 \\ 0 & 15 & 6 \\ 0 & -25 & -10 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = 0$$

$$R_2 \rightarrow \frac{R_2}{3}, R_3 \rightarrow -\frac{1}{5}R_3$$

$$\begin{bmatrix} 1 & 10 & 5 \\ 0 & 5 & 2 \\ 0 & 5 & 2 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = 0$$

$$R_3 \rightarrow R_3 - R_2$$

$$\begin{bmatrix} 1 & 10 & 5 \\ 0 & 5 & 2 \\ 0 & 0 & 0 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = 0$$

This shows that the solution is one linearly independent

$$\Rightarrow \begin{aligned} x_1 + 10x_2 + 5x_3 &= 0 \\ 5x_2 + 2x_3 &= 0 \end{aligned}$$

$$\frac{x_1}{-5} = -\frac{x_2}{2} = \frac{x_3}{5}$$

or

$$\frac{x_1}{5} = \frac{x_2}{2} = -\frac{x_3}{5}$$

Taking proportional values, $x_1 = 5$, $x_2 = 2$, $x_3 = -5$

$$\therefore X_2 = \begin{bmatrix} 5 \\ 2 \\ -5 \end{bmatrix}$$

And the third eigen vector corresponding to repeated root $\lambda_3 = 2 = \lambda_2$ will be same because the solution is not two linearly independent. Hence the vectors X_2 and X_3 are not linearly independent. Similarity transformation is not possible.

$$\text{Therefore } \lambda = 2, 2, 3 \text{ and } X_1 = \begin{bmatrix} 1 \\ 1 \\ -2 \end{bmatrix}, X_2 = X_3 = \begin{bmatrix} 5 \\ 2 \\ -5 \end{bmatrix}.$$

Example 8. Verify the statement that the sum of the elements in the diagonal of a matrix is the sum of the eigen values of the matrix.

$$A = \begin{bmatrix} -2 & 2 & -3 \\ 2 & 1 & -6 \\ -1 & -2 & 0 \end{bmatrix}.$$

Sol. The characteristic equation is $|A - \lambda I| = 0$

$$\Rightarrow \begin{vmatrix} -2-\lambda & 2 & -3 \\ 2 & 1-\lambda & -6 \\ -1 & -2 & -\lambda \end{vmatrix} = 0$$

$$\begin{aligned}
&\Rightarrow (-2 - \lambda) [(1 - \lambda)(-\lambda) - 12] - 2(-2\lambda - 6) - 3[-4 + (1 - \lambda)] = 0 \\
&\Rightarrow (-2 - \lambda) (\lambda^2 - \lambda - 12) + 4(\lambda + 3) + 3(\lambda + 3) = 0 \\
&\Rightarrow -2\lambda^2 + 2\lambda + 24 - \lambda^3 + \lambda^2 + 12\lambda + 4\lambda + 12 + 3\lambda + 9 = 0 \\
&\Rightarrow -\lambda^3 - \lambda^2 + 21\lambda + 45 = 0 \\
&\Rightarrow \lambda^3 + \lambda^2 - 21\lambda - 45 = 0.
\end{aligned}$$

This is cubic equation in λ and hence it has 3 roots *i.e.*, there are three eigen values.

The sum of the eigen values = $-\left(\frac{\text{coefficient of } \lambda^2}{\text{coefficient of } \lambda^3}\right) = -1$.

The sum of the elements on the diagonal of the matrix A
 $= -2 + 1 + 0 = -1$. Hence the result.

Example 9. Find the eigen values and corresponding eigen vectors of the matrix,

$$A = \begin{bmatrix} -5 & 2 \\ 2 & -2 \end{bmatrix} \quad (\text{U.P.T.U., 2008})$$

Sol. The characteristic equation is $|A - \lambda I| = 0$

$$\begin{aligned}
&\Rightarrow \begin{vmatrix} -5-\lambda & 2 \\ 2 & -2-\lambda \end{vmatrix} = (5 + \lambda)(2 + \lambda) - 4 = 0 \\
&\Rightarrow \lambda^2 + 7\lambda + 6 = 0 \Rightarrow (\lambda + 6)(\lambda + 1) = 0 \\
&\therefore \lambda = -1, -6
\end{aligned}$$

Now, we consider the relation $(A - \lambda I)X = 0$...(i)

For $\lambda = -1$

$$\begin{aligned}
&\begin{bmatrix} -5+1 & 2 \\ 2 & -2+1 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = 0 \\
&\Rightarrow \begin{bmatrix} -4 & 2 \\ 2 & -1 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = 0
\end{aligned}$$

$R_2 \rightarrow R_2 + 2R_1$, on coefficient matrix

$$\begin{aligned}
&\begin{bmatrix} -4 & 2 \\ 0 & 0 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = 0 \\
&\Rightarrow -2x_1 + x_2 = 0 \quad \text{...(ii)}
\end{aligned}$$

Let $x_2 = k$

$$\therefore \text{From (i)} \quad x_1 = \frac{k}{2}$$

$$X_1 = \begin{bmatrix} k/2 \\ k \end{bmatrix} = \frac{k}{2} \begin{bmatrix} 1 \\ 2 \end{bmatrix} \text{ or } \begin{bmatrix} 1 \\ 2 \end{bmatrix}$$

For $\lambda = -6$, from (i), we get

$$\begin{bmatrix} 1 & 2 \\ 2 & 4 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = 0$$

$R_2 \rightarrow R_2 - 2R_1$, on coefficient matrix

$$\begin{bmatrix} 1 & 2 \\ 0 & 0 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = 0$$

$$\Rightarrow x_1 + 2x_2 = 0 \quad \dots(iii)$$

Let $x_2 = k \quad \therefore x_1 = -2k$

$$X_2 = \begin{bmatrix} -2k \\ k \end{bmatrix} = k \begin{bmatrix} -2 \\ 1 \end{bmatrix} \text{ or } \begin{bmatrix} -2 \\ 1 \end{bmatrix}$$

Hence, the eigen values are $\lambda = -1, -6$

And eigen vectors are $X_1 = \begin{bmatrix} 1 \\ 2 \end{bmatrix}$

$$X_2 = \begin{bmatrix} -2 \\ 1 \end{bmatrix}.$$

EXERCISE 3.6

Find the eigen values and eigen vectors of:

1. $\begin{bmatrix} 1 & 2 \\ 1 & 0 \end{bmatrix}$ $\left[\text{Ans. } 2, -1, \begin{bmatrix} 2 \\ 1 \end{bmatrix}, \begin{bmatrix} 1 \\ -1 \end{bmatrix} \right]$

2. $\begin{bmatrix} 6 & 8 \\ 8 & -6 \end{bmatrix}$ $\left[\text{Ans. } 10, -10, \begin{bmatrix} 2 \\ 1 \end{bmatrix}, \begin{bmatrix} 1 \\ -2 \end{bmatrix} \right]$

3. $\begin{bmatrix} 1 & 1 & 3 \\ 1 & 5 & 1 \\ 3 & 1 & 1 \end{bmatrix}$ $\left[\text{Ans. } -2, 3, 6, \begin{bmatrix} -2 \\ 0 \\ 2 \end{bmatrix}, \begin{bmatrix} -1 \\ 1 \\ -1 \end{bmatrix}, \begin{bmatrix} 1 \\ 2 \\ 1 \end{bmatrix} \right]$

4. $\begin{bmatrix} 8 & -6 & 2 \\ -6 & 7 & -4 \\ 2 & -4 & 3 \end{bmatrix}$ $\left[\text{Ans. } 0, 3, 15, \begin{bmatrix} 1 \\ 2 \\ 2 \end{bmatrix}, \begin{bmatrix} 2 \\ 1 \\ -2 \end{bmatrix}, \begin{bmatrix} 2 \\ -2 \\ 1 \end{bmatrix} \right]$

5. $\begin{bmatrix} -2 & 1 & 1 \\ -11 & 4 & 5 \\ -1 & 1 & 0 \end{bmatrix}$ $\left[\text{Ans. } -1, 1, 2, \begin{bmatrix} 0 \\ -1 \\ 1 \end{bmatrix}, \begin{bmatrix} 1 \\ 2 \\ 1 \end{bmatrix}, \begin{bmatrix} 1 \\ 2 \\ 1 \end{bmatrix} \right]$

6. $\begin{bmatrix} 2 & 1 & 0 \\ 0 & 2 & 1 \\ 0 & 0 & 2 \end{bmatrix}$ $\left[\text{Ans. } 2, 2, 2, \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}, \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}, \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix} \right]$

7. $\begin{bmatrix} 1 & -1 & -1 \\ 1 & -1 & 0 \\ 1 & 0 & -1 \end{bmatrix}$ $\left[\text{Ans. } -1, \pm i, \begin{bmatrix} 0 \\ 1 \\ -1 \end{bmatrix}, \begin{bmatrix} 1+i \\ 1 \\ 1 \end{bmatrix}, \begin{bmatrix} 1-i \\ 1 \\ 1 \end{bmatrix} \right]$

$$8. \begin{bmatrix} 3 & 1 & 4 \\ 0 & 2 & 0 \\ 0 & 0 & 5 \end{bmatrix}$$

$$\left[\text{Ans. } 2, 3, 5, \begin{bmatrix} 0 \\ -1 \\ 0 \end{bmatrix}, \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}, \begin{bmatrix} 3 \\ 2 \\ 1 \end{bmatrix} \right]$$

$$9. \begin{bmatrix} -2 & 2 & -3 \\ 2 & 1 & -6 \\ -1 & -2 & 0 \end{bmatrix}$$

$$\left[\text{Ans. } 5, -3, -3, \begin{bmatrix} -1 \\ -2 \\ 1 \end{bmatrix}, \begin{bmatrix} -2 \\ 1 \\ 0 \end{bmatrix}, \begin{bmatrix} 3 \\ 0 \\ 1 \end{bmatrix} \right]$$

$$10. \begin{bmatrix} 3 & -2 & -5 \\ 4 & -1 & -5 \\ -2 & -1 & -3 \end{bmatrix}$$

$$\left[\text{Ans. } 5, 2, 2, \begin{bmatrix} 3 \\ 2 \\ 4 \end{bmatrix}, \begin{bmatrix} 1 \\ 3 \\ -1 \end{bmatrix}, \begin{bmatrix} 1 \\ 3 \\ -1 \end{bmatrix} \right]$$

$$11. \begin{bmatrix} 1 & 1 & 3 \\ 1 & 5 & 1 \\ 3 & 1 & 1 \end{bmatrix}$$

$$\left[\text{Ans. } -2, 3, 6, \begin{bmatrix} -1 \\ 0 \\ 1 \end{bmatrix}, \begin{bmatrix} 1 \\ -1 \\ 1 \end{bmatrix}, \begin{bmatrix} 1 \\ 2 \\ 1 \end{bmatrix} \right]$$

$$12. \begin{bmatrix} 4 & -20 & -10 \\ -2 & 10 & 4 \\ 6 & -30 & -13 \end{bmatrix}$$

$$\left[\text{Ans. } 0, -1, 2, \begin{bmatrix} 5 \\ 1 \\ 0 \end{bmatrix}, \begin{bmatrix} 2 \\ 0 \\ 1 \end{bmatrix}, \begin{bmatrix} 0 \\ 1 \\ -2 \end{bmatrix} \right]$$

$$13. \begin{bmatrix} -1 & 0 & 2 \\ 0 & 1 & 2 \\ 2 & 2 & 0 \end{bmatrix}$$

$$\left[\text{Ans. } 0, 3, -3, \begin{bmatrix} 2 \\ -2 \\ 1 \end{bmatrix}, \begin{bmatrix} 1 \\ 2 \\ 2 \end{bmatrix}, \begin{bmatrix} 2 \\ 1 \\ -2 \end{bmatrix} \right]$$

$$14. \begin{bmatrix} 11 & -4 & -7 \\ 7 & -2 & -5 \\ 10 & -4 & -6 \end{bmatrix}$$

$$\left[\text{Ans. } 0, 1, 2, \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix}, \begin{bmatrix} 1 \\ -1 \\ 2 \end{bmatrix}, \begin{bmatrix} 2 \\ 1 \\ 2 \end{bmatrix} \right]$$

$$15. \begin{bmatrix} -9 & 4 & 4 \\ -8 & 3 & 4 \\ -16 & 8 & 7 \end{bmatrix}$$

$$\left[\text{Ans. } -1, -1, 3, \begin{bmatrix} 0 \\ 1 \\ -1 \end{bmatrix}, \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix}, \begin{bmatrix} 1 \\ 1 \\ 2 \end{bmatrix} \right]$$

$$16. \begin{bmatrix} 2 & 4 & 6 \\ 4 & 2 & -6 \\ -6 & -6 & -15 \end{bmatrix}$$

$$\left[\text{Ans. } -2, 9, -18, \begin{bmatrix} 1 \\ -1 \\ 0 \end{bmatrix}, \begin{bmatrix} 2 \\ 2 \\ -1 \end{bmatrix}, \begin{bmatrix} 1 \\ 1 \\ 4 \end{bmatrix} \right]$$

$$17. \begin{bmatrix} 1 & 2 & 3 \\ 2 & 4 & 6 \\ 3 & 6 & 9 \end{bmatrix}$$

$$\left[\text{Ans. } 0, 0, 14, \begin{bmatrix} -2 \\ 1 \\ 0 \end{bmatrix}, \begin{bmatrix} 3 \\ 6 \\ -5 \end{bmatrix}, \begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix} \right]$$

$$18. \begin{bmatrix} 1 & -4 & -1 & -4 \\ 2 & 0 & 5 & -4 \\ -1 & 1 & -2 & 3 \\ -1 & 4 & -1 & 6 \end{bmatrix}$$

$$[\text{Ans. } \lambda^4 - 5\lambda^3 + 9\lambda^2 - 7\lambda + 2 = 0, \lambda = 2, 1, 1, 1]$$

$$\begin{bmatrix} 2 \\ 3 \\ -2 \\ -3 \end{bmatrix}, \begin{bmatrix} 3 \\ 6 \\ -4 \\ -5 \end{bmatrix}$$

19. $\begin{bmatrix} 2 & -2 & 2 \\ 1 & 1 & 1 \\ 1 & 3 & -1 \end{bmatrix}$.

Ans. $2, 2, -2, \begin{bmatrix} 0 \\ 1 \\ 1 \end{bmatrix}, \begin{bmatrix} -4 \\ 1 \\ 7 \end{bmatrix}, \begin{bmatrix} -4 \\ 1 \\ 7 \end{bmatrix}$

20. Find the eigen values of A^5 when

$$A = \begin{bmatrix} 3 & 0 & 0 \\ 5 & 4 & 0 \\ 3 & 6 & 1 \end{bmatrix}.$$

[Ans. $3^5, 4^5, 1^5$]

21. If A and B are $n \times n$ matrices and B is a non-singular matrix prove that A and $B^{-1}AB$ have same eigen values.

[Hint : Characteristic polynomial of $B^{-1}AB = |B^{-1}AB - \lambda I| = |B^{-1}AB - B^{-1}(\lambda I)B|$
 $= |B^{-1}(A - \lambda I)B| = |B^{-1}| |A - \lambda I| |B| = |B^{-1}| |B| |A - \lambda I| = |B^{-1}B| |A - \lambda I|$
 $= |I| |A - \lambda I| = |A - \lambda I| = \text{characteristic polynomial of } A].$

22. Find the sum and product of the eigen values.

$$A = \begin{bmatrix} 2 & 3 & -2 \\ -2 & 1 & 1 \\ 1 & 0 & 2 \end{bmatrix}.$$

[Ans. Sum = $2 + 1 + 2 = 5$, product = $|A| = 21$]

3.18 CAYLEY-HAMILTON THEOREM

STATEMENT

Any square matrix A satisfies its own characteristic equation.

(U.P.T.U., 2005)

i.e., if $C_0 + C_1\lambda + C_2\lambda^2 + \dots + C_n\lambda^n$ is the characteristic polynomial of degree n in λ then $C_0I + C_1A + C_2A^2 + \dots + C_nA^n = 0$.

Proof: Let A be a square matrix of order n .

$$\text{Let } |A - \lambda I| = C_0 + C_1\lambda + C_2\lambda^2 + \dots + C_n\lambda^n \quad \dots(i)$$

be the characteristic polynomial of A .

Now, the elements of $\text{adj}(A - \lambda I)$ are cofactors of the matrix $(A - \lambda I)$ and are polynomials in λ of degree not exceeding $n - 1$.

$$\text{Thus } \text{adj}(A - \lambda I) = B_0 + B_1\lambda + B_2\lambda^2 + \dots + B_{n-1}\lambda^{n-1} \quad \dots(ii)$$

where B_i are n -square matrices whose elements are the functions of the elements of A and independent of λ .

$$\text{We know that } (A - \lambda I) \cdot \text{adj}(A - \lambda I) = |A - \lambda I| I$$

From (i) and (ii), we have

$$(A - \lambda I) (B_0 + B_1\lambda + B_2\lambda^2 + \dots + B_{n-1}\lambda^{n-1}) = (C_0 + C_1\lambda + C_2\lambda^2 + \dots + C_n\lambda^n)I \quad \dots(iii)$$

Equating the like powers of λ on both sides of (iii), we get

$$\begin{aligned} AB_0 &= C_0I \\ AB_1 - B_0 &= C_1I \end{aligned}$$

$$\begin{array}{ccc} \vdots & \vdots & \vdots \\ AB_{n-1} - B_{n-2} & = & C_{n-1}I \\ -B_{n-1} & = & C_n I \end{array}$$

Pre-multiplying the above equations by $I, A, A^2, \dots, A^n$ respectively and adding; we get

$$C_0 I + C_1 A + C_2 A^2 + \dots + C_n A^n = 0 \quad \dots(iv)$$

Since all the terms on the L.H.S. cancel each other. Thus A satisfies its own characteristic equation. **Proved.**

3.18.1 Inverse by Cayley-Hamilton Theorem

$$\text{We have } C_0 I + C_1 A + C_2 A^2 + \dots + C_n A^n = 0 \quad \dots(i)$$

Multiplying equation (i) by A^{-1} , we get

$$\begin{aligned} C_0 A^{-1} + C_1 I + C_2 A + \dots + C_n A^{n-1} &= 0 \\ \Rightarrow C_0 A^{-1} &= -[C_1 I + C_2 A + \dots + C_n A^{n-1}] \end{aligned}$$

$$\Rightarrow A^{-1} = -\frac{1}{C_0} [C_1 I + C_2 A + \dots + C_n A^{n-1}]$$

Note: A^{-1} exists only if $C_0 = |A|$.

Definition. An expression of the form $C_0 + C_1 \lambda + C_2 \lambda^2 + \dots + C_n \lambda^n$ where $C_0, C_1, \dots, C_n$ are square matrices of the same order and $C_n \neq 0$ is called a “matrix polynomial of degrees n ”.

Example 1. Verify the Cayley-Hamilton theorem for the matrix $\begin{bmatrix} 1 & 2 & 3 \\ 2 & 4 & 5 \\ 3 & 5 & 6 \end{bmatrix}$. Also find its

inverse using this theorem.

(U.P.T.U., 2006)

Sol. The characteristic equation of A is

$$|A - \lambda I| = \begin{vmatrix} 1-\lambda & 2 & 3 \\ 2 & 4-\lambda & 5 \\ 3 & 5 & 6-\lambda \end{vmatrix} = 0$$

$$\Rightarrow (1 - \lambda) [(4 - \lambda)(6 - \lambda) - 25] - 2[2(6 - \lambda) - 15] + 3[10 - 3(4 - \lambda)] = 0$$

$$\Rightarrow \lambda^3 - 11\lambda^2 - 4\lambda + 1 = 0$$

Cayley-Hamilton theorem is verified if A satisfies the above characteristic equation i.e.

$$A^3 - 11A^2 - 4A + 1 = 0 \quad \dots(i)$$

$$\text{Now, } A^2 = A \cdot A = \begin{bmatrix} 1 & 2 & 3 \\ 2 & 4 & 5 \\ 3 & 5 & 6 \end{bmatrix} \begin{bmatrix} 1 & 2 & 3 \\ 2 & 4 & 5 \\ 3 & 5 & 6 \end{bmatrix} = \begin{bmatrix} 14 & 25 & 31 \\ 25 & 45 & 56 \\ 31 & 56 & 70 \end{bmatrix}$$

$$A^3 = A \cdot A^2 = \begin{bmatrix} 1 & 2 & 3 \\ 2 & 4 & 5 \\ 3 & 5 & 6 \end{bmatrix} \begin{bmatrix} 14 & 25 & 31 \\ 25 & 45 & 56 \\ 31 & 56 & 70 \end{bmatrix} = \begin{bmatrix} 157 & 283 & 353 \\ 283 & 510 & 636 \\ 353 & 636 & 793 \end{bmatrix}$$

$$\text{Verification, } A^3 - 11A^2 - 4A + I = \begin{bmatrix} 157 & 283 & 353 \\ 283 & 510 & 636 \\ 353 & 636 & 793 \end{bmatrix} - 11 \begin{bmatrix} 14 & 25 & 31 \\ 25 & 45 & 56 \\ 31 & 56 & 70 \end{bmatrix} - \begin{bmatrix} 1 & 2 & 3 \\ 2 & 4 & 5 \\ 3 & 5 & 6 \end{bmatrix} + \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} = \begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}$$

$$\Rightarrow A^3 - 11A^2 - 4A + I = 0$$

Hence theorem verified.

To find A^{-1} : Multiplying equation by A^{-1} , we get

$$A^2 - 11A - 4I + A^{-1} = 0$$

$$\Rightarrow A^{-1} = -A^2 + 11A + 4I$$

$$\begin{aligned} \Rightarrow A^{-1} &= - \begin{bmatrix} 14 & 25 & 31 \\ 25 & 45 & 56 \\ 31 & 56 & 70 \end{bmatrix} + 11 \begin{bmatrix} 1 & 2 & 3 \\ 2 & 4 & 5 \\ 3 & 5 & 6 \end{bmatrix} + 4 \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \\ &= \begin{bmatrix} 1 & -3 & 2 \\ -3 & 3 & -1 \\ 2 & -1 & 0 \end{bmatrix}. \end{aligned}$$

Example 2. Express $B = A^8 - 11A^7 - 4A^6 + A^5 + A^4 - 11A^3 - 3A^2 + 2A + I$ as a quadratic polynomial in A , where A is square matrix given in Example 1. Find B as well as A^4 .

$$\begin{aligned} \text{Sol. } B &= A^8 - 11A^7 - 4A^6 + A^5 + A^4 - 11A^3 - 3A^2 + 2A + I \\ &= A^5 (A^3 - 11A^2 - 4A + I) + A (A^3 - 11A^2 - 4A + I) + A^2 + A + I \\ &= A^5 (0) + A(0) + A^2 + A + I \quad \text{As } A^3 - 11A^2 - 4A + I = 0 \text{ in Example 1} \\ &= A^2 + A + I \end{aligned}$$

$$\text{Thus, } B = \begin{bmatrix} 14 & 25 & 31 \\ 25 & 45 & 56 \\ 31 & 56 & 70 \end{bmatrix} + \begin{bmatrix} 1 & 2 & 3 \\ 2 & 4 & 5 \\ 3 & 5 & 6 \end{bmatrix} + \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

$$B = \begin{bmatrix} 16 & 27 & 34 \\ 27 & 50 & 61 \\ 34 & 61 & 77 \end{bmatrix}.$$

To find A^4 : We have

$$A^3 - 11A^2 - 4A + I = 0$$

$$\Rightarrow A^3 = 11A^2 + 4A - I$$

Multiplying both sides by A

$$A^4 = 11A^3 + 4A^2 - IA$$

$$\begin{aligned} A^4 &= 11 \begin{bmatrix} 157 & 283 & 353 \\ 283 & 510 & 636 \\ 353 & 636 & 793 \end{bmatrix} + 4 \begin{bmatrix} 14 & 25 & 31 \\ 25 & 45 & 56 \\ 31 & 56 & 70 \end{bmatrix} - \begin{bmatrix} 1 & 2 & 3 \\ 2 & 4 & 5 \\ 3 & 5 & 6 \end{bmatrix} \\ &= \begin{bmatrix} 1782 & 3211 & 4004 \\ 3211 & 5786 & 7215 \\ 4004 & 7215 & 8997 \end{bmatrix}. \end{aligned}$$

Example 3. Find the characteristic equation of the matrix $\begin{bmatrix} 2 & -1 & 1 \\ -1 & 2 & -1 \\ 1 & -2 & 2 \end{bmatrix}$ and hence also find

A^{-1} by Cayley-Hamilton theorem.

(U.P.T.U., 2003, 2004, 2005)

Sol. The characteristic equation of A is

$$|A - \lambda I| = \begin{vmatrix} 2-\lambda & -1 & 1 \\ -1 & 2-\lambda & -1 \\ 1 & -2 & 2-\lambda \end{vmatrix} = 0$$

$$\Rightarrow (2 - \lambda) [\lambda^2 - 4\lambda + 4 - 2] + [-2 + \lambda + 1] + [2 - 2 + \lambda] = 0$$

$$\Rightarrow \lambda^3 - 6\lambda^2 + 8\lambda - 3 = 0$$

By Cayley-Hamilton theorem $A^3 - 6A^2 + 8A - 3I = 0$.

Multiplying by A^{-1} on both sides, we get

$$A^2 - 6A + 8I - 3A^{-1} = 0$$

$$\Rightarrow A^{-1} = \frac{1}{3} (A^2 - 6A + 8I)$$

$$\text{Now, } A^2 = A.A = \begin{bmatrix} 2 & -1 & 1 \\ -1 & 2 & -1 \\ 1 & -2 & 2 \end{bmatrix} \begin{bmatrix} 2 & -1 & 1 \\ -1 & 2 & -1 \\ 1 & -2 & 2 \end{bmatrix} = \begin{bmatrix} 6 & -6 & 6 \\ -5 & 7 & -5 \\ 6 & -9 & 7 \end{bmatrix}$$

$$\therefore A^2 - 6A + 8I = \begin{bmatrix} 6 & -6 & 6 \\ -5 & 7 & -5 \\ 6 & -9 & 7 \end{bmatrix} - 6 \begin{bmatrix} 2 & -1 & 1 \\ -1 & 2 & -1 \\ 1 & -2 & 2 \end{bmatrix} + \begin{bmatrix} 8 & 0 & 0 \\ 0 & 8 & 0 \\ 0 & 0 & 8 \end{bmatrix} = \begin{bmatrix} 2 & 0 & -1 \\ 1 & 3 & 1 \\ 0 & 3 & 3 \end{bmatrix}$$

$$\text{Thus, } A^{-1} = \frac{1}{3} \begin{bmatrix} 2 & 0 & -1 \\ 1 & 3 & 1 \\ 0 & 3 & 3 \end{bmatrix}.$$

Example 4. Evaluate A^{-1} , A^{-2} , A^{-3} if

$$A = \begin{bmatrix} 4 & 6 & 6 \\ 1 & 3 & 2 \\ -1 & -4 & -3 \end{bmatrix}$$

Sol. Characteristic equation is

$$|A - \lambda I| = \begin{vmatrix} 4-\lambda & 6 & 6 \\ 1 & 3-\lambda & 2 \\ -1 & -4 & -3-\lambda \end{vmatrix} = 0$$

$$\Rightarrow \lambda^3 - 4\lambda^2 - \lambda + 4 = 0$$

By Cayley-Hamilton theorem, we have

$$A^3 - 4A^2 - A + 4I = 0$$

$$\Rightarrow A^2 - 4A - I + 4A^{-1} = 0 \quad (\text{multiplying by } A^{-1})$$

$$A^{-1} = \frac{1}{4} [-A^2 + 4A + I] \quad \dots(i)$$

Now,
$$A^2 = \begin{bmatrix} 16 & 18 & 18 \\ 5 & 7 & 6 \\ -5 & -6 & -5 \end{bmatrix}$$

$$\begin{aligned} \therefore -A^2 + 4A + I &= \begin{bmatrix} -16 & -18 & -18 \\ -5 & -7 & -6 \\ 5 & 6 & 5 \end{bmatrix} + \begin{bmatrix} 16 & 24 & 24 \\ 4 & 12 & 8 \\ -4 & -16 & -12 \end{bmatrix} + \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \\ &= \begin{bmatrix} 1 & 6 & 6 \\ -1 & 6 & 2 \\ 1 & -10 & -6 \end{bmatrix} \end{aligned}$$

Thus,
$$A^{-1} = \frac{1}{4} \begin{bmatrix} 1 & 6 & 6 \\ -1 & 6 & 2 \\ 1 & -10 & -6 \end{bmatrix}$$

Multiplying equation (1) by A^{-1} , we get

$$A^{-2} = \frac{1}{4} [-A + 4I + IA^{-1}] = \frac{1}{4} \begin{bmatrix} 1/4 & -1/2 & -9/2 \\ -5/4 & 5/2 & -3/2 \\ 5/4 & 3/2 & 11/2 \end{bmatrix}$$

Similarly,
$$\begin{aligned} A^{-3} &= \frac{1}{4} [-I + 4A^{-1} + A^{-2}] \\ &= \frac{1}{64} \begin{bmatrix} 1 & 78 & 78 \\ -21 & 90 & 26 \\ 21 & -154 & -90 \end{bmatrix}. \end{aligned}$$

Example 5. Verify Cayley-Hamilton theorem for the matrix $A = \begin{bmatrix} 1 & 2 \\ 2 & -1 \end{bmatrix}$ and hence find A^{-1} . (U.P.T.U., 2008)

Sol. The characteristic equation of A is

$$|A - \lambda I| = \begin{vmatrix} 1-\lambda & 2 \\ 2 & -1-\lambda \end{vmatrix} = 0 \Rightarrow -(1 - \lambda^2) - 4 = 0$$

or
$$\lambda^2 - 5 = 0$$

By Cayley-Hamilton theorem $A^2 - 5I = 0$...(i)

Now
$$A^2 = \begin{bmatrix} 1 & 2 \\ 2 & -1 \end{bmatrix} \begin{bmatrix} 1 & 2 \\ 2 & -1 \end{bmatrix} = \begin{bmatrix} 5 & 0 \\ 0 & 5 \end{bmatrix} = 5 \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} = 5I$$

Putting value of A^2 in equation (i), we get

$$A^2 - 5I = 5I - 5I = 0$$

Hence, the Cayley-Hamilton theorem is verified.

To find A^{-1} : Multiplying equation (i) by A^{-1} , we get

$$A - 5A^{-1} = 0 \Rightarrow A^{-1} = \frac{1}{5}A = \frac{1}{5} \begin{bmatrix} 1 & 2 \\ 2 & -1 \end{bmatrix}.$$

Example 6. Show that the matrix $A = \begin{bmatrix} 0 & c & -b \\ -c & 0 & a \\ b & -a & 0 \end{bmatrix}$ satisfies its characteristic equation. Also find A^{-1} .

Sol. $|A - \lambda I| = \begin{vmatrix} -\lambda & c & -b \\ -c & -\lambda & a \\ b & -a & -\lambda \end{vmatrix} = 0$

i.e., $-\lambda[\lambda^2 + a^2] - c[\lambda c - ab] - b[ac + b\lambda] = 0$

i.e., $\lambda^3 + \lambda(a^2 + b^2 + c^2) = 0$

By Cayley-Hamilton theorem, $A^3 + A(a^2 + b^2 + c^2) = 0$... (i)

Now, $A^2 = \begin{bmatrix} 0 & c & -b \\ -c & 0 & a \\ b & -a & 0 \end{bmatrix} \begin{bmatrix} 0 & c & -b \\ -c & 0 & a \\ b & -a & 0 \end{bmatrix} = \begin{bmatrix} -(c^2 + b^2) & ab & ac \\ ab & -(c^2 + a^2) & bc \\ ac & bc & -(b^2 + a^2) \end{bmatrix}$

$$A^3 = A \cdot A^2 = \begin{bmatrix} 0 & c & -b \\ -c & 0 & a \\ b & -a & 0 \end{bmatrix} \begin{bmatrix} -(b^2 + c^2) & ab & ac \\ ab & -(c^2 + a^2) & bc \\ ac & bc & -(b^2 + a^2) \end{bmatrix}$$

$$\text{or } A^3 = \begin{bmatrix} 0 & -c(a^2 + b^2 + c^2) & b(a^2 + b^2 + c^2) \\ c(a^2 + b^2 + c^2) & 0 & -a(a^2 + b^2 + c^2) \\ -b(a^2 + b^2 + c^2) & a(a^2 + b^2 + c^2) & 0 \end{bmatrix} = -(a^2 + b^2 + c^2) \begin{bmatrix} 0 & c & -b \\ -c & 0 & a \\ b & -a & 0 \end{bmatrix}$$

or $A^3 = -(a^2 + b^2 + c^2) A$... (ii)

Using (ii) in equation (i), we get

$$A^3 + A(a^2 + b^2 + c^2) = -(a^2 + b^2 + c^2) A + A(a^2 + b^2 + c^2) = 0$$

To find A^{-1} : Multiplying equation (i) A^{-2} on both sides, we get

$$A + A^{-1}(a^2 + b^2 + c^2) = 0$$

$$\Rightarrow A^{-1} = -\frac{1}{(a^2 + b^2 + c^2)} \cdot A$$

Hence, $A^{-1} = -\frac{1}{(a^2 + b^2 + c^2)} \begin{bmatrix} 0 & c & -b \\ -c & 0 & a \\ b & -a & 0 \end{bmatrix}.$

EXERCISE 3.7

1. Find the characteristic equation of the matrix A

$$A = \begin{bmatrix} 4 & 3 & 1 \\ 2 & 1 & -2 \\ 1 & 2 & 1 \end{bmatrix}. \text{ Hence find } A^{-1}. \quad (\text{U.P.T.U., 2001})$$

$$\left[\text{Ans. Characteristic equation : } \lambda^3 - 6\lambda^2 + 6\lambda - 11 = 0, A^{-1} = \frac{1}{11} \begin{bmatrix} 5 & -1 & -7 \\ -4 & 3 & 10 \\ 3 & -5 & -2 \end{bmatrix} \right]$$

2. Find the characteristic equation of the matrix

$$A = \begin{bmatrix} 2 & 1 & 1 \\ 0 & 1 & 0 \\ 1 & 1 & 2 \end{bmatrix}. \text{ Verify Cayley-Hamilton theorem and hence} \quad (\text{U.P.T.U., 2002})$$

evaluate the matrix equation $A^8 - 5A^7 + 7A^6 - 3A^5 + A^4 - 5A^3 + 8A^2 - 2A + I$.

$$\left[\text{Ans. } \lambda^3 - 5\lambda^2 + 7\lambda - 3 = 0, A^2 + A + I, \begin{bmatrix} 8 & 5 & 5 \\ 0 & 3 & 0 \\ 5 & 5 & 8 \end{bmatrix} \right]$$

3. Verify Cayley-Hamilton theorem for A and hence find A^{-1}

$$(i) \begin{bmatrix} 2 & -1 & 1 \\ -15 & 6 & -5 \\ 5 & -2 & 2 \end{bmatrix} \quad \left[\text{Ans. } A^{-1} = \begin{bmatrix} -2 & 0 & 1 \\ -5 & 1 & 5 \\ 0 & 1 & 3 \end{bmatrix} \right]$$

$$(ii) \begin{bmatrix} 1 & 1 & 1 \\ 1 & 2 & 1 \\ 1 & 1 & 2 \end{bmatrix} \quad \left[\text{Ans. } A^{-1} = \begin{bmatrix} 3 & -1 & -1 \\ -1 & 1 & 0 \\ -1 & 0 & 1 \end{bmatrix} \right]$$

$$(iii) \begin{bmatrix} 1 & 0 & 3 \\ 2 & 1 & -1 \\ 1 & -1 & 1 \end{bmatrix} \quad \left[\text{Ans. } A^{-1} = -\frac{1}{9} \begin{bmatrix} 0 & -3 & -3 \\ -3 & -2 & 7 \\ -3 & 1 & 1 \end{bmatrix} \right]$$

$$4. \text{ Find } A^{-1} \text{ if } A = \begin{bmatrix} 7 & 2 & -2 \\ -6 & -1 & 2 \\ 6 & 2 & -1 \end{bmatrix}. \quad \left[\text{Ans. } A^{-1} = \frac{1}{3} \begin{bmatrix} -3 & -2 & 2 \\ 6 & 5 & -2 \\ -6 & -2 & 5 \end{bmatrix} \right]$$

$$5. \text{ Show that the matrix } A = \begin{bmatrix} 1 & 2 & 0 \\ 2 & -1 & 0 \\ 0 & 0 & -1 \end{bmatrix} \text{ satisfies its own characteristic equation and hence}$$

or otherwise obtain the value of A^{-1} .

$$\left[\text{Ans. } A^{-1} = \begin{bmatrix} 1/5 & 0 & 0 \\ 0 & 1/5 & 0 \\ 0 & 0 & 1 \end{bmatrix} \right]$$

6. Find A^{-1} if $A = \begin{bmatrix} 1 & 2 & 3 \\ 1 & 3 & 5 \\ 1 & 5 & 12 \end{bmatrix}$.

Ans. $A^{-1} = \frac{1}{3} \begin{bmatrix} 11 & -9 & 1 \\ -7 & 9 & -2 \\ 2 & -3 & 1 \end{bmatrix}$

7. If $A = \begin{bmatrix} 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 1 & 0 \\ 1 & 0 & 0 & 0 \end{bmatrix}$ find A^{-1} .

Ans. $A^{-1} = \begin{bmatrix} 0 & 0 & 0 & 1 \\ 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 1 & 0 & 0 \end{bmatrix}$

8. Find $B = A^6 - 4A^5 + 8A^4 - 12A^3 + 14A^2$ if $A = \begin{bmatrix} 1 & 2 \\ -1 & 3 \end{bmatrix}$.

Ans. $\lambda^2 - 4\lambda + 5 = 0, B = 5I - 4A = \begin{bmatrix} 1 & -8 \\ 4 & -7 \end{bmatrix}$

9. Find A^{-1} if $A = \begin{bmatrix} 3 & 3 & 4 \\ 2 & -3 & 4 \\ 0 & -1 & 1 \end{bmatrix}$.

Ans. $A^{-1} = \frac{1}{11} \begin{bmatrix} -1 & 7 & -24 \\ 2 & -3 & 4 \\ 2 & -3 & 15 \end{bmatrix}$

10. Show that the matrix $A = \begin{bmatrix} 2 & -3 & 1 \\ 3 & 1 & 3 \\ -5 & 2 & -4 \end{bmatrix}$ satisfies the equation $A(A - I)(A + 2I) = 0$.

3.19 DIAGONALIZATION OF A MATRIX

A square matrix 'A' is said to be diagonalisable if there exists another non-singular square matrix P such that $P^{-1}AP$ is a diagonal matrix. In other words, A is diagonalisable if A is similar to a diagonal matrix.

Similar matrix: Two matrices A and B are said to be similar if there exists a non-singular matrix P such that $B = P^{-1}AP$.

This transformation of A to B is known as similarity transformation.

Note: Similar matrices A and B have same eigen values. Further, if X is an eigen vector of A then $Y = P^{-1}X$ is an eigen vector of the matrix B .

3.19.1 Working Rule

The matrix P is called a model matrix for A . There are following steps to obtain diagonal form of matrix A .

- Step 1:** First of all obtain eigen values $\lambda_1, \lambda_2, \dots, \lambda_n$ of the matrix A .
- Step 2:** If there exists a pair of distinct complex eigen values then A is not diagonalisable over the field R of real numbers.
- Step 3:** If there does not exist a set of n linearly independent eigen vectors, then A is not diagonalisable.
- Step 4:** If there exist n linearly independent eigen vectors.

$$30. A = \begin{bmatrix} 2 & 1 & -3 & -6 \\ 3 & -3 & 1 & 2 \\ 1 & 1 & 1 & 2 \end{bmatrix}. \quad (\text{U.P.T.U., 2002})$$

$$\left[\text{Ans. } P = \begin{bmatrix} 0 & 0 & 1 \\ -1 & 0 & \frac{2}{9} \\ \frac{-3}{14} & \frac{1}{28} & \frac{2}{28} \end{bmatrix}, Q = \begin{bmatrix} 1 & -1 & 4 & 0 \\ 0 & 1 & -5 & 0 \\ 0 & 0 & 1 & -2 \\ 0 & 0 & 0 & 1 \end{bmatrix} \text{ and rank} = 3 \right]$$

$$31. A = \begin{bmatrix} 1 & 1 & 2 \\ 1 & 2 & 3 \\ 0 & -1 & -1 \end{bmatrix}.$$

$$\left[\text{Ans. } P = \begin{bmatrix} 1 & 0 & 0 \\ -1 & 1 & 0 \\ -1 & 1 & 1 \end{bmatrix}, Q = \begin{bmatrix} 1 & -1 & -1 \\ 0 & 1 & -1 \\ 0 & 0 & 1 \end{bmatrix} \right]$$

$$32. A = \begin{bmatrix} 1 & 2 & 3 & -2 \\ 2 & -2 & 1 & 3 \\ 3 & 0 & 4 & 1 \end{bmatrix}. \quad \left[\text{Ans. } P = \begin{bmatrix} 1 & 0 & 0 \\ -2 & 1 & 0 \\ -1 & -1 & 1 \end{bmatrix}, Q = \begin{bmatrix} 1 & 1/3 & -4/3 & -\frac{1}{3} \\ 0 & -1/6 & -5/6 & 7/6 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \right] \text{rank} = 2$$

3.13 SYSTEM OF LINEAR EQUATIONS (NON-HOMOGENEOUS)

Let us consider the following system of m linear equations in n unknowns $x_1, x_2, \dots, x_n$:

$$\left. \begin{aligned} a_{11}x_1 + a_{12}x_2 + \dots + a_{1n}x_n &= b_1 \\ a_{21}x_1 + a_{22}x_2 + \dots + a_{2n}x_n &= b_2 \\ \dots & \dots \dots = \dots \\ a_{m1}x_1 + a_{m2}x_2 + \dots + a_{mn}x_n &= b_n \end{aligned} \right\} \quad \dots(i)$$

In matrix notation these equations can be put in the form

$$AX = B \quad \dots(ii)$$

where

$$A = \begin{bmatrix} a_{11} & a_{12} & \dots & a_{1n} \\ a_{21} & a_{22} & \dots & a_{2n} \\ \dots & \dots & \dots & \dots \\ a_{m1} & a_{m2} & \dots & a_{mn} \end{bmatrix}$$

$$X = \begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{bmatrix} \text{ and } B = \begin{bmatrix} b_1 \\ b_2 \\ \vdots \\ b_n \end{bmatrix}$$

Augmented matrix: The augmented matrix $[A : B]$ or $\tilde{A}$ of system (i) is obtained by augmenting A by the column B .

$$\text{i.e.,} \quad \tilde{A} = [A : B] = \begin{bmatrix} a_{11} & a_{12} & \cdots & a_{1n} & \vdots & b_1 \\ a_{21} & a_{22} & \cdots & a_{2n} & \vdots & b_2 \\ \cdots & \cdots & \cdots & \cdots & \vdots & \cdots \\ a_{m1} & a_{m2} & & a_{mn} & \vdots & b_n \end{bmatrix}$$

3.13.1 Conditions for Solution of Linear Equations [U.P.T.U., (C.O.), 2003]

Consistent: If the ranks of A and augmented matrix $[A : B]$ are equal, then the system is said to be consistent otherwise inconsistent. There are following conditions for exist the solution of any system of linear equations :

(i) If $\rho(A) = \rho[A : B] = r = n$ (where n is the number of variables) then the system has a unique solution.

(ii) If $\rho(A) = \rho[A : B] = r < n$.

then the system has infinitely many solutions in terms of remaining $n - r$ unknowns which are arbitrary.

If $n - r = 1$ (then solution is one variable independent solution and let equal to K).
 $n - r = 2$ (then solution is two variable independent solution and let variables equal to K_1 , and K_2) and so on.

Trivial solution: It is a solution where all x_i are zero i.e., $x_1 = x_2 = \cdots = x_n = 0$.

Example 1. Check the consistency of the following system of linear nonhomogeneous equations and find the solution, if exists: (U.P.T.U., 2007)

$$7x_1 + 2x_2 + 3x_3 = 16$$

$$2x_1 + 11x_2 + 5x_3 = 25$$

$$x_1 + 3x_2 + 4x_3 = 13.$$

Sol. Here,

$$A = \begin{bmatrix} 7 & 2 & 3 \\ 2 & 11 & 5 \\ 1 & 3 & 4 \end{bmatrix}, X = \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix}, B = \begin{bmatrix} 16 \\ 25 \\ 13 \end{bmatrix}$$

$$\text{The augmented matrix } [A : B] = \begin{bmatrix} 7 & 2 & 3 & \vdots & 16 \\ 2 & 11 & 5 & \vdots & 25 \\ 1 & 3 & 4 & \vdots & 13 \end{bmatrix}$$

Applying $R_1 \leftrightarrow R_3$, we get

$$[A : B] \sim \begin{bmatrix} 1 & 3 & 4 & \vdots & 13 \\ 2 & 11 & 5 & \vdots & 25 \\ 7 & 2 & 2 & \vdots & 16 \end{bmatrix}$$

Again $R_2 \rightarrow R_2 - 2R_1$, $R_3 \rightarrow R_3 - 7R_1$

$$\sim \begin{bmatrix} 1 & 3 & 4 & \vdots & 13 \\ 0 & 5 & -3 & \vdots & -1 \\ 0 & -19 & -25 & \vdots & -75 \end{bmatrix}$$

$$R_3 \rightarrow R_3 + \frac{19}{5} R_2$$

$$\sim \begin{bmatrix} 1 & 3 & 4 & \vdots & 13 \\ 0 & 5 & -3 & \vdots & -1 \\ 0 & 0 & \frac{-182}{5} & \vdots & -\frac{394}{5} \end{bmatrix}$$

$$\Rightarrow \rho(A) = \rho[A : B] = 3$$

$$\Rightarrow r = n = 3. \therefore \text{The system is consistent.}$$

The given system has a unique solution.

$$\text{Now from } AX = B$$

$$\begin{bmatrix} 1 & 3 & 4 \\ 0 & 5 & -3 \\ 0 & 0 & \frac{-182}{5} \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 13 \\ -1 \\ \frac{-394}{5} \end{bmatrix}$$

$$\Rightarrow \begin{bmatrix} x_1 + 3x_2 + 4x_3 \\ 5x_2 - 3x_3 \\ \frac{-182}{5}x_3 \end{bmatrix} = \begin{bmatrix} 13 \\ -1 \\ \frac{-394}{5} \end{bmatrix}$$

$$\Rightarrow x_1 + 3x_2 + 4x_3 = 13 \quad \dots(i)$$

$$5x_2 - 3x_3 = -1 \quad \dots(ii)$$

$$\frac{182}{5}x_3 = \frac{394}{5} \quad \dots(iii)$$

On solving these equations, we get the final solution

$$x_1 = \frac{95}{91}, x_2 = \frac{100}{91}, x_3 = \frac{197}{91}.$$

Example 2. Test the consistency of following system of linear equations and hence find the solution. (U.P.T.U., 2005)

$$\begin{aligned} 4x_1 - x_2 &= 12 \\ -x_1 + 5x_2 - 2x_3 &= 0 \\ -2x_2 + 4x_3 &= -8 \end{aligned}$$

$$\text{Sol. The augmented matrix } [A : B] = \begin{bmatrix} 4 & -1 & 0 & \vdots & 12 \\ -1 & 5 & -2 & \vdots & 0 \\ 0 & -2 & 4 & \vdots & -8 \end{bmatrix}$$

Applying $R_1 \leftrightarrow R_2$, we get

$$[A : B] \sim \begin{bmatrix} -1 & 5 & -2 & \vdots & 0 \\ 4 & -1 & 0 & \vdots & 12 \\ 0 & -2 & 4 & \vdots & -8 \end{bmatrix}$$

$$R_2 \rightarrow R_2 + 4R_1$$

$$\sim \begin{bmatrix} -1 & 5 & -2 & \vdots & 0 \\ 0 & 19 & -8 & \vdots & 12 \\ 0 & -2 & 4 & \vdots & -8 \end{bmatrix}$$

$$R_1 \rightarrow -R_1, R_3 \rightarrow R_3 + \frac{2}{19} R_2,$$

$$\sim \begin{bmatrix} 1 & -5 & 2 & \vdots & 0 \\ 0 & 19 & -8 & \vdots & 12 \\ 0 & 0 & \frac{60}{19} & \vdots & \frac{-128}{19} \end{bmatrix}$$

$$\therefore \rho(A) = \rho[A : B] = 3$$

$$\text{i.e., } r = n = 3 \text{ (The system is consistent).}$$

Hence, there is a unique solution

$$\Rightarrow \begin{bmatrix} -1 & -5 & 2 \\ 0 & 19 & -8 \\ 0 & 0 & \frac{60}{19} \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 0 \\ 12 \\ \frac{-128}{19} \end{bmatrix}$$

$$\Rightarrow x_1 - 5x_2 + 2x_3 = 0 \quad \dots(i)$$

$$19x_2 - 8x_3 = 12 \quad \dots(ii)$$

$$\frac{60x_3}{19} = \frac{-128}{19} \quad \dots(iii)$$

$$\Rightarrow x_3 = -\frac{32}{15}$$

putting the value of x_3 in equation (ii), we get

$$19x_2 - 8\left(-\frac{32}{15}\right) = 12$$

$$\Rightarrow 19x_2 = 12 - \frac{256}{15} = -\frac{76}{15}$$

$$\Rightarrow x_2 = \frac{-76}{15 \times 19} = -\frac{4}{15}$$

and putting the values of x_1, x_2 in equation (i), we get

$$x_1 - 5\left(-\frac{4}{15}\right) + 2\left(-\frac{32}{15}\right) = 0$$

$$x_1 + \frac{20}{15} - \frac{64}{15} = 0$$

$$\Rightarrow x_1 - \frac{44}{15} = 0$$

$$\Rightarrow x_1 = \frac{44}{15}$$

$$\text{Hence, } x_1 = \frac{44}{15}, x_2 = -\frac{4}{15} \text{ and } x_3 = -\frac{32}{15}.$$

Example 3. Solve

$$2x_1 - 2x_2 + 4x_3 + 3x_4 = 9$$

$$x_1 - x_2 + 2x_3 + 2x_4 = 6$$

$$2x_1 - 2x_2 + x_3 + 2x_4 = 3$$

$$x_1 - x_2 + x_4 = 2$$

Sol. The augmented matrix is

$$[A : B] = \begin{bmatrix} 2 & -2 & 4 & 3 & : & 9 \\ 1 & -1 & 2 & 2 & : & 6 \\ 2 & -2 & 1 & 2 & : & 3 \\ 1 & -1 & 0 & 1 & : & 2 \end{bmatrix}$$

$$R_1 \leftrightarrow R_2$$

$$\sim \begin{bmatrix} 1 & -1 & 2 & 2 & : & 6 \\ 2 & -2 & 4 & 3 & : & 9 \\ 2 & -2 & 1 & 2 & : & 3 \\ 1 & -1 & 0 & 1 & : & 2 \end{bmatrix}$$

$$R_2 \rightarrow R_2 - 2R_1, R_4 \rightarrow R_4 - R_1, R_3 \rightarrow R_3 - 2R_1$$

$$\sim \begin{bmatrix} 1 & -1 & 2 & 2 & : & 6 \\ 0 & 0 & 0 & -1 & : & -3 \\ 0 & 0 & -3 & -2 & : & -9 \\ 0 & 0 & -2 & -1 & : & -4 \end{bmatrix}$$

$$R_2 \rightarrow (-1) R_2, R_3 \rightarrow (-1) R_3, R_4 \rightarrow (-1) R_4$$

$$\sim \begin{bmatrix} 1 & -1 & 2 & 2 & : & 6 \\ 0 & 0 & 0 & 1 & : & 3 \\ 0 & 0 & 3 & 2 & : & 9 \\ 0 & 0 & 2 & 1 & : & 4 \end{bmatrix}$$

$$R_3 \rightarrow R_3 - R_4$$

$$\sim \begin{bmatrix} 1 & -1 & 2 & 2 & : & 6 \\ 0 & 0 & 0 & 1 & : & 3 \\ 0 & 0 & 1 & 1 & : & 5 \\ 0 & 0 & 2 & 1 & : & 4 \end{bmatrix}$$

$$R_3 \leftrightarrow R_2$$

$$\sim \begin{bmatrix} 1 & -1 & 2 & 2 & : & 6 \\ 0 & 0 & 1 & 1 & : & 5 \\ 0 & 0 & 0 & 1 & : & 3 \\ 0 & 0 & 2 & 1 & : & 4 \end{bmatrix}$$

$$R_4 \leftrightarrow R_3$$

$$\sim \begin{bmatrix} 1 & -1 & 2 & 2 & \vdots & 6 \\ 0 & 0 & 1 & 1 & \vdots & 5 \\ 0 & 0 & 2 & 1 & \vdots & 4 \\ 0 & 0 & 0 & 1 & \vdots & 3 \end{bmatrix}$$

$$R_3 \rightarrow R_3 - 2R_2$$

$$\sim \begin{bmatrix} 1 & -1 & 2 & 2 & \vdots & 6 \\ 0 & 0 & 1 & 1 & \vdots & 5 \\ 0 & 0 & 0 & -1 & \vdots & -6 \\ 0 & 0 & 0 & 1 & \vdots & 3 \end{bmatrix} \sim \begin{bmatrix} 1 & -1 & 2 & 2 & \vdots & 6 \\ 0 & 0 & 1 & 1 & \vdots & 5 \\ 0 & 0 & 0 & 1 & \vdots & 6 \\ 0 & 0 & 0 & 1 & \vdots & 3 \end{bmatrix} (R_3 \rightarrow -R_3)$$

$$R_4 \rightarrow R_4 - R_3 \text{ and then } R_4 \rightarrow (-1) R_4$$

$$\sim \begin{bmatrix} 1 & -1 & 2 & 2 & \vdots & 6 \\ 0 & 0 & 1 & 1 & \vdots & 5 \\ 0 & 0 & 0 & 1 & \vdots & 6 \\ 0 & 0 & 0 & 0 & \vdots & 3 \end{bmatrix}$$

Hence, $\rho(A) = 3$ and $\rho[A : B] = 4 \Rightarrow \rho(A) \neq \rho[A : B]$.

So the given system is inconsistent and therefore it has no solution.

Example 4. Investigate for what values of λ, μ the equations

$$x + y + z = 6, \quad x + 2y + 3z = 10, \quad x + 2y + \lambda z = \mu$$

have (i) no solution (ii) a unique solution (iii) an infinity of solutions.

(U.P.T.U., 2001)

Sol. The augmented matrix

$$[A : B] = \begin{bmatrix} 1 & 1 & 1 & \vdots & 6 \\ 1 & 2 & 3 & \vdots & 10 \\ 1 & 2 & \lambda & \vdots & \mu \end{bmatrix}$$

$$R_2 \rightarrow R_2 - R_1, \quad R_3 \rightarrow R_3 - R_1$$

$$\sim \begin{bmatrix} 1 & 1 & 1 & \vdots & 6 \\ 0 & 1 & 2 & \vdots & 4 \\ 0 & 1 & \lambda - 1 & \vdots & \mu - 6 \end{bmatrix}$$

$$R_3 \rightarrow R_3 - R_2$$

$$\sim \begin{bmatrix} 1 & 1 & 1 & \vdots & 6 \\ 0 & 1 & 2 & \vdots & 4 \\ 0 & 0 & \lambda - 3 & \vdots & \mu - 10 \end{bmatrix}$$

(i) For no solution $\rho(A) \neq \rho[A : B]$ it is only possible when $\lambda = 3$.

(ii) For unique solution $\rho(A) = \rho[A : B]$ it is only possible when $\lambda - 3 \neq 0$ i.e., $\lambda \neq 3$ and $\mu \neq 10$.

(iii) For infinite number of solutions $\rho(A) = \rho[A : B] = r < n$ it is only possible when $\lambda = 3$ and $\mu = 10$.

Example 5. Show that the equations

$$\begin{aligned}x + y + z &= 6 \\x + 2y + 3z &= 14 \\x + 4y + 7z &= 30\end{aligned}$$

are consistent and solve them.

Sol. The augmented matrix is

$$[A : B] = \begin{bmatrix} 1 & 1 & 1 & : & 6 \\ 1 & 2 & 3 & : & 14 \\ 1 & 4 & 7 & : & 30 \end{bmatrix}$$

$$R_2 \rightarrow R_2 - R_1, R_3 \rightarrow R_3 - R_1$$

$$\sim \begin{bmatrix} 1 & 1 & 1 & : & 6 \\ 0 & 1 & 2 & : & 8 \\ 0 & 3 & 6 & : & 24 \end{bmatrix}$$

$$R_3 \rightarrow R_3 - 3R_2$$

$$\sim \begin{bmatrix} 1 & 1 & 1 & : & 6 \\ 0 & 1 & 2 & : & 8 \\ 0 & 0 & 0 & : & 0 \end{bmatrix}$$

Hence,

$$\rho(A) = \rho[A : B] = 2$$

i.e.,

$$r = 2 < 3 \quad (n = 3)$$

$\therefore$

$$n - r = 3 - 2 = 1 \text{ (one variable independent solution).}$$

The system is consistent and have infinitely solutions.

Now

$$AX = B$$

$$\therefore \begin{bmatrix} 1 & 1 & 1 \\ 0 & 1 & 2 \\ 0 & 0 & 0 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 6 \\ 8 \\ 0 \end{bmatrix}$$

$$x + y + z = 6 \quad \dots(i)$$

$$y + 2z = 8 \quad \dots(ii)$$

Let

$$z = k$$

Putting $z = k$ in (ii), we get

$$y + 2k = 8 \Rightarrow y = 8 - 2k$$

$$\text{From (i)} \quad x + 8 - 2k + k = 6 \Rightarrow x = k - 2$$

Therefore, $x = k - 2$, $y = 8 - 2k$ and $z = k$. **Ans.**

Example 6. Solve

$$3x + 3y + 2z = 1$$

$$x + 2y = 4$$

$$10y + 3z = -2$$

$$2x - 3y - z = 5$$

Sol. The augmented matrix is

$$[A : B] = \begin{bmatrix} 3 & 3 & 2 & : & 1 \\ 1 & 2 & 0 & : & 4 \\ 0 & 10 & 3 & : & -2 \\ 2 & -3 & -1 & : & 5 \end{bmatrix}$$

$$R_1 \leftrightarrow R_3$$

$$\sim \begin{bmatrix} 1 & 2 & 0 & : & 4 \\ 3 & 3 & 2 & : & 1 \\ 0 & 10 & 3 & : & -2 \\ 2 & -3 & -1 & : & 5 \end{bmatrix}$$

$$R_2 \rightarrow R_2 - 3R_1, R_4 \rightarrow R_4 - 2R_1$$

$$\sim \begin{bmatrix} 1 & 2 & 0 & : & 4 \\ 0 & -3 & 2 & : & -11 \\ 0 & 10 & 3 & : & -2 \\ 0 & -7 & -1 & : & -3 \end{bmatrix}$$

$$R_3 \rightarrow R_3 + \frac{10}{3}R_2, R_4 \rightarrow R_4 - \frac{7}{3}R_2$$

$$\sim \begin{bmatrix} 1 & 2 & 0 & : & 4 \\ 0 & -3 & 2 & : & -11 \\ 0 & 0 & \frac{29}{3} & : & \frac{-116}{3} \\ 0 & 0 & \frac{-17}{3} & : & \frac{68}{3} \end{bmatrix}$$

$$R_3 \rightarrow \frac{3}{29}R_3, R_4 \rightarrow \frac{3}{17}R_4$$

$$\sim \begin{bmatrix} 1 & 2 & 0 & : & 4 \\ 0 & -3 & 2 & : & -11 \\ 0 & 0 & 1 & : & -4 \\ 0 & 0 & -1 & : & 4 \end{bmatrix}$$

$$R_4 \rightarrow R_4 + R_3$$

$$\sim \begin{bmatrix} 1 & 2 & 0 & : & 4 \\ 0 & -3 & 2 & : & -11 \\ 0 & 0 & 1 & : & -4 \\ 0 & 0 & 0 & : & 0 \end{bmatrix}$$

$$\Rightarrow \rho(A) = \rho[A : B] = 3$$

i.e., $r = 3 = n = \text{number of variables.}$

Hence, the system is consistent and has unique solution.

Now, $AX = B$

$$\Rightarrow \begin{bmatrix} 1 & 2 & 0 \\ 0 & -3 & 2 \\ 0 & 0 & 1 \\ 0 & 0 & 0 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 4 \\ -11 \\ -4 \\ 0 \end{bmatrix}$$

$$\Rightarrow \begin{aligned} x + 2y &= 4 & \dots(i) \end{aligned}$$

$$\begin{aligned} -3y + 2z &= -11 & \dots(ii) \end{aligned}$$

$$\begin{aligned} z &= -4 & \dots(iii) \end{aligned}$$

On solving (i) and (ii), we get $x = 2$, $y = 1$. Hence, $x = 2$, $y = 1$ and $z = -4$.

Example 7. Apply the matrix method to solve the system of equations

$$x + 2y - z = 3$$

$$3x - y + 2z = 1$$

$$2x - 2y + 3z = 2$$

$$x - y + z = -1$$

[U.P.T.U., 2003; U.P.T.U. (C.O.), 2003]

Sol. The augmented matrix is

$$[A : B] = \begin{bmatrix} 1 & 2 & -1 & : & 3 \\ 3 & -1 & 2 & : & 1 \\ 2 & -2 & 3 & : & 2 \\ 1 & -1 & 1 & : & -1 \end{bmatrix}$$

$$R_2 \rightarrow R_2 - 3R_1, R_3 \rightarrow R_3 - 2R_1, R_4 \rightarrow R_4 - R_1$$

$$\sim \begin{bmatrix} 1 & 2 & -1 & : & 3 \\ 0 & -7 & 5 & : & -8 \\ 0 & -6 & 5 & : & -4 \\ 0 & -3 & 2 & : & -4 \end{bmatrix}$$

$$R_3 \rightarrow R_3 - \frac{6}{7}R_2, R_4 \rightarrow R_4 - \frac{3}{7}R_2$$

$$\sim \begin{bmatrix} 1 & 2 & -1 & : & 3 \\ 0 & -7 & 5 & : & -8 \\ 0 & 0 & \frac{5}{7} & : & \frac{20}{7} \\ 0 & 0 & -\frac{1}{7} & : & -\frac{4}{7} \end{bmatrix}$$

$$R_4 \rightarrow R_4 + \frac{1}{5}R_3$$

$$\sim \begin{bmatrix} 1 & 2 & -1 & : & 3 \\ 0 & -7 & 5 & : & -8 \\ 0 & 0 & \frac{5}{7} & : & \frac{20}{7} \\ 0 & 0 & 0 & & 0 \end{bmatrix}$$

$$\Rightarrow \rho(A) = \rho[A : B] = 3 = \text{number of variables.}$$

Hence, the system is consistent and has a unique solution.

$$\text{Now, } AX = B$$

$$\Rightarrow \begin{bmatrix} 1 & 2 & -1 \\ 0 & -7 & 5 \\ 0 & 0 & \frac{5}{7} \\ 0 & 0 & 0 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 3 \\ -8 \\ \frac{20}{7} \\ 0 \end{bmatrix}$$

$$\Rightarrow \begin{aligned} x + 2y - z &= 3 & \dots(i) \\ -7y + 5z &= -8 & \dots(ii) \end{aligned}$$

$$\frac{5}{7}z = \frac{20}{7} \Rightarrow z = 4$$

From (i) and (ii), we get, $x = -1, y = 4$

$$\Rightarrow x = -1, y = 4, z = 4.$$

3.14 SYSTEM OF HOMOGENEOUS EQUATIONS

If in the set of equations (1) of (3.13), $b_1 = b_2 = \dots = b_n = 0$, the set of equation is said to be homogeneous.

Result 1: If $r = n$, i.e., the rank of coefficient matrix is equal to the number of variables, then there is always a trivial solution ($x_1 = x_2 = \dots = x_n = 0$).

Result 2: If $r < n$, i.e., the rank of coefficient matrix is smaller than the number of variables, then there exist a non-trivial solution.

Result 3: For non-trivial solution always $|A| = 0$.

Example 8. Solve the following system of homogeneous equations:

$$\begin{aligned} x + 2y + 3z &= 0 \\ 3x + 4y + 4z &= 0 \\ 7x + 10y + 12z &= 0 \end{aligned}$$

Sol. Here,

$$A = \begin{bmatrix} 1 & 2 & 3 \\ 3 & 4 & 4 \\ 7 & 10 & 12 \end{bmatrix}$$

$$\sim \begin{bmatrix} 1 & 2 & 3 \\ 0 & -2 & -8 \\ 0 & -4 & -9 \end{bmatrix}$$

$$R_3 \rightarrow R_3 - 2R_2$$

$$A \sim \begin{bmatrix} 1 & 2 & 3 \\ 0 & -2 & -5 \\ 0 & 0 & 1 \end{bmatrix}$$

This shows $\text{rank}(A) = 3 = \text{number of unknowns}$. Hence, the given system has a trivial solution *i.e.*, $x = y = z = 0$.

Example 9. Solve

$$x + y - 2z + 3w = 0$$

$$x - 2y + z - w = 0$$

$$4x + y - 5z + 8w = 0$$

$$5x - 7y + 2z - w = 0$$

Sol. The coefficient matrix A is

$$A = \begin{bmatrix} 1 & 1 & -2 & 3 \\ 1 & -2 & 1 & -1 \\ 4 & 1 & -5 & 8 \\ 5 & -7 & 2 & -1 \end{bmatrix}$$

$$R_2 \rightarrow R_2 - R_1, R_3 \rightarrow R_3 - 4R_1, R_4 \rightarrow R_4 - 5R_1$$

$$\sim \begin{bmatrix} 1 & 1 & -2 & 3 \\ 0 & -3 & 3 & -4 \\ 0 & -3 & 3 & -4 \\ 0 & -12 & 12 & -16 \end{bmatrix}$$

$$R_3 \rightarrow R_3 - R_2, R_4 \rightarrow R_4 - 4R_2$$

$$\sim \begin{bmatrix} 1 & 1 & -2 & 3 \\ 0 & -3 & 3 & -4 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix}$$

$\Rightarrow \rho(A) = 2 < 4$ ($n = 4$), so there exist a non-trivial solution.

Now,

$$AX = B$$

$$\Rightarrow \begin{bmatrix} 1 & 1 & -2 & 3 \\ 0 & -3 & 3 & -4 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \\ w \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \\ 0 \end{bmatrix}$$

$$\Rightarrow \quad x + y - 2z + 3w = 0 \quad \dots(i)$$

$$\quad - 3y + 3z - 4w = 0 \quad \dots(ii)$$

Choose $z = k_1$, $w = k_2$, then from (ii) and (i), we get

$$- 3y + 3k_1 - 4k_2 = 0 \Rightarrow y = k_1 - \frac{4}{3}k_2$$

and $x + k_1 - \frac{4}{3}k_2 - 2k_1 + 3k_2 = 0 \Rightarrow x = k_1 - \frac{5}{3}k_2$.

where k_1 and k_2 are arbitrary constants.

Example 10. Find the value of λ such that the following equations have unique solution.

$$\lambda x + 2y - 2z - 1 = 0, \quad 4x + 2\lambda y - z - 2 = 0, \quad 6x + 6y + \lambda z - 3 = 0 \quad (U.P.T.U., 2003)$$

Sol. We have

$$\lambda x + 2y - 2z = 1$$

$$4x + 2\lambda y - z = 2$$

$$6x + 6y + \lambda z = 3$$

The coefficient matrix A is

$$A = \begin{bmatrix} \lambda & 2 & -2 \\ 4 & 2\lambda & -1 \\ 6 & 6 & \lambda \end{bmatrix}$$

For unique solution $|A| \neq 0$

$$\therefore \quad \begin{vmatrix} \lambda & 2 & -2 \\ 4 & 2\lambda & -1 \\ 6 & 6 & \lambda \end{vmatrix} = \lambda^3 + 11\lambda - 30 \neq 0$$

$$\Rightarrow (\lambda - 2)(\lambda^2 + 2\lambda + 15) \neq 0 \Rightarrow \lambda \neq 2.$$

Example 11. Determine b such that the system of homogeneous equations (U.P.T.U., 2008)

$$2x + y + 2z = 0$$

$$x + y + 3z = 0$$

$$4x + 3y + bz = 0$$

has (i) trivial solution and (ii) non-trivial solution.

Sol. (i) For trivial solution, $|A| \neq 0$

$$\therefore \quad |A| = \begin{vmatrix} 2 & 1 & 2 \\ 1 & 1 & 3 \\ 4 & 3 & b \end{vmatrix} \neq 0$$

$$\Rightarrow 2(b - 9) - (b - 12) + 2(3 - 4) \neq 0$$

$$\Rightarrow 2b - 18 - b + 12 - 2 \neq 0 \Rightarrow b - 8 \neq 0 \Rightarrow b \neq 8.$$

(ii) For non-trivial solution, $|A| = 0$

$$\Rightarrow b - 8 = 0 \Rightarrow b = 8.$$

3.15 GAUSSIAN ELIMINATION METHOD

Gaussian elimination method is an exact method which solves a given system of equations in n unknowns by transforming the coefficient matrix into an upper triangular matrix and then solve for the unknowns by back substitution.

Example 12. Solve the system of equations:

$$2x_1 + 3x_2 + x_3 = 9$$

$$x_1 + 2x_2 + 3x_3 = 6$$

$$3x_1 + x_2 + 2x_3 = 8$$

(U.P.T.U., 2006)

by Gaussian elimination method.

Sol. The augmented matrix is:

$$[A : B] = \begin{bmatrix} 2 & 3 & 1 & : & 9 \\ 1 & 2 & 3 & : & 6 \\ 3 & 1 & 2 & : & 8 \end{bmatrix}$$

$$R_1 \leftrightarrow R_2$$

$$\sim \begin{bmatrix} 1 & 2 & 3 & : & 6 \\ 2 & 3 & 1 & : & 9 \\ 3 & 1 & 2 & : & 8 \end{bmatrix}$$

$$R_2 \rightarrow R_2 - 2R_1, R_3 \rightarrow R_3 - 3R_1$$

$$\sim \begin{bmatrix} 1 & 2 & 3 & : & 6 \\ 0 & -1 & -5 & : & -3 \\ 0 & -5 & -7 & : & -10 \end{bmatrix}$$

$$R_3 \rightarrow R_3 - 5R_2$$

$$\sim \begin{bmatrix} 1 & 2 & 3 & : & 6 \\ 0 & -1 & -5 & : & -3 \\ 0 & 0 & 18 & : & 5 \end{bmatrix}$$

which is upper triangular form

$$\therefore 18x_3 = 5 \Rightarrow x_3 = \frac{5}{18}$$

$$\text{and } x_1 + 2x_2 + 3x_3 = 6 \quad \dots(i)$$

$$-x_2 - 5x_3 = -3 \quad \dots(ii)$$

$$\text{From (ii)} \quad x_2 + \frac{25}{18} = 3 \Rightarrow x_2 = 3 - \frac{25}{18} = \frac{29}{18}$$

again from (i), we have

$$x_1 + 2 \times \frac{29}{18} + 3 \times \frac{5}{18} = 6 \Rightarrow x_1 + \frac{73}{18} = 6$$

$$\Rightarrow x_1 = 6 - \frac{73}{18} = \frac{35}{18}$$

$$\text{Hence, } x_1 = \frac{35}{18}, x_2 = \frac{29}{18} \text{ and } x_3 = \frac{5}{18}.$$

Example 13. Solve by Gaussian elimination method

$$10x_1 - 7x_2 + 3x_3 + 5x_4 = 6$$

$$-6x_1 + 8x_2 - x_3 - 4x_4 = 5$$

$$3x_1 + x_2 + 4x_3 + 11x_4 = 2$$

$$5x_1 - 9x_2 - 2x_3 + 4x_4 = 7$$

Sol. The augmented matrix is

$$[A : B] = \begin{bmatrix} 10 & -7 & 3 & 5 & : & 6 \\ -6 & 8 & -1 & -4 & : & 5 \\ 3 & 1 & 4 & 11 & : & 2 \\ 5 & -9 & -2 & 4 & : & 7 \end{bmatrix}$$

$$R_3 \leftrightarrow R_2$$

$$\sim \begin{bmatrix} 10 & -7 & 3 & 5 & : & 6 \\ 3 & 1 & 4 & 11 & : & 2 \\ -6 & 8 & -1 & -4 & : & 5 \\ 5 & -9 & -2 & 4 & : & 7 \end{bmatrix}$$

$$R_2 \rightarrow R_2 - \frac{3}{10}R_1, R_3 \rightarrow R_3 + \frac{6}{10}R_1, R_4 \rightarrow R_4 - \frac{1}{2}R_1$$

$$\sim \begin{bmatrix} 10 & -7 & 3 & 5 & : & 6 \\ 0 & \frac{31}{10} & \frac{31}{10} & \frac{19}{2} & : & \frac{1}{5} \\ 0 & \frac{19}{5} & \frac{4}{5} & -1 & : & \frac{43}{5} \\ 0 & -\frac{11}{2} & -\frac{7}{2} & \frac{3}{2} & : & 4 \end{bmatrix}$$

$$R_2 \rightarrow \frac{10}{31}R_2, R_3 \rightarrow \frac{5}{19}R_2, R_3 \rightarrow -\frac{2}{11}R_3$$

$$\sim \begin{bmatrix} 10 & -7 & 3 & 5 & : & 6 \\ 0 & 1 & 1 & \frac{95}{31} & : & \frac{2}{31} \\ 0 & 1 & \frac{4}{19} & -\frac{5}{19} & : & \frac{43}{19} \\ 0 & 1 & \frac{7}{11} & -\frac{3}{11} & : & \frac{-8}{11} \end{bmatrix}$$

$$R_3 \rightarrow R_3 - R_2, R_4 \rightarrow R_4 - R_2$$

$$\sim \begin{bmatrix} 10 & -7 & 3 & 5 & : & 6 \\ 0 & 1 & 1 & \frac{95}{31} & : & \frac{2}{31} \\ 0 & 0 & -\frac{15}{19} & -\frac{1960}{589} & : & \frac{1295}{589} \\ 0 & 0 & -\frac{4}{11} & -\frac{1138}{341} & : & -\frac{270}{341} \end{bmatrix}$$

$$R_3 \rightarrow -\frac{19}{15}R_3, R_4 \rightarrow -\frac{11}{4}R_4$$

$$\sim \begin{bmatrix} 10 & -7 & 3 & 5 & : & 6 \\ 0 & 1 & 1 & \frac{95}{31} & : & \frac{2}{31} \\ 0 & 0 & 1 & \frac{392}{93} & : & -\frac{259}{93} \\ 0 & 0 & 1 & \frac{569}{62} & : & \frac{135}{62} \end{bmatrix}$$

$$R_4 \rightarrow R_4 - R_3$$

$$\sim \begin{bmatrix} 10 & -7 & 3 & 5 & : & 6 \\ 0 & 1 & 1 & \frac{95}{31} & : & \frac{2}{31} \\ 0 & 0 & 1 & \frac{392}{93} & : & -\frac{259}{93} \\ 0 & 0 & 0 & \frac{923}{186} & : & \frac{923}{186} \end{bmatrix}$$

Hence, the coefficient matrix is an upper triangular form

$$\therefore \frac{923}{186}x_4 = \frac{923}{186} \Rightarrow x_4 = 1$$

$$x_3 + \frac{392}{93}x_4 = -\frac{259}{93} \Rightarrow x_3 + \frac{392}{93} = -\frac{259}{93}$$

or
$$x_3 = -\frac{259}{93} - \frac{392}{93} = -\frac{651}{93} = -7$$

and
$$x_2 + x_3 + \frac{95}{31}x_4 = \frac{2}{31} \Rightarrow x_2 - 7 + \frac{95}{31} = \frac{2}{31}$$

$$\Rightarrow x_2 - \frac{122}{31} = \frac{2}{31} \Rightarrow x_2 = \frac{2}{31} + \frac{122}{31} = \frac{124}{31} = 4$$

Again $10x_1 - 7x_2 + 3x_3 + 5x_4 = 6$

$$\Rightarrow 10x_1 - 7 \times 4 + 3 \times (-7) + 5 \times 1 = 6 \Rightarrow 10x_1 - 28 - 21 + 5 = 6$$

$$\Rightarrow 10x_1 - 44 = 6 \Rightarrow 10x_1 = 50 \Rightarrow x_1 = 5$$

Therefore, $x_1 = 5$, $x_2 = 4$, $x_3 = -7$ and $x_4 = 1$.

3.15.1 Gauss-Jordan Elimination Method

Apply elementary row operations on both A and B such that A reduces to the normal form. Then the solution is obtained.

Example 14. Solve by Gauss-Jordan elimination method:

$$2x_1 + x_2 + 3x_3 = 1$$

$$4x_1 + 4x_2 + 7x_3 = 1$$

$$2x_1 + 5x_2 + 9x_3 = 3$$

Sol. Here $A = \begin{bmatrix} 2 & 1 & 3 \\ 4 & 4 & 7 \\ 2 & 5 & 9 \end{bmatrix}$, $B = \begin{bmatrix} 1 \\ 1 \\ 3 \end{bmatrix}$

$\therefore$ Augmented matrix

$$[A : B] = \begin{bmatrix} 2 & 1 & 3 & : & 1 \\ 4 & 4 & 7 & : & 1 \\ 2 & 5 & 9 & : & 3 \end{bmatrix}$$

$$R_2 \rightarrow R_2 - 2R_1, R_3 \rightarrow R_3 - R_1$$

$$\sim \begin{bmatrix} 2 & 1 & 3 & : & 1 \\ 0 & 2 & 1 & : & -1 \\ 0 & 4 & 6 & : & 2 \end{bmatrix}$$

$$R_3 \rightarrow R_3 - 2R_2$$

$$\sim \begin{bmatrix} 2 & 1 & 3 & : & 1 \\ 0 & 2 & 1 & : & -1 \\ 0 & 0 & 4 & : & 4 \end{bmatrix}$$

$$R_1 \rightarrow \frac{1}{2}R_1, R_2 \rightarrow \frac{1}{2}R_2, R_3 \rightarrow \frac{1}{4}R_3$$

$$\sim \begin{bmatrix} 1 & \frac{1}{2} & \frac{3}{2} & : & \frac{1}{2} \\ 0 & 1 & \frac{1}{2} & : & -\frac{1}{2} \\ 0 & 0 & 1 & : & 1 \end{bmatrix}$$

$$R_1 \rightarrow R_1 - \frac{1}{2}R_2$$

$$\sim \begin{bmatrix} 1 & 0 & \frac{5}{4} & : & \frac{3}{4} \\ 0 & 1 & \frac{1}{2} & : & -\frac{1}{2} \\ 0 & 0 & 1 & : & 1 \end{bmatrix}$$

$$R_1 \rightarrow R_1 - \frac{5}{4}R_3, R_2 \rightarrow R_2 - \frac{1}{2}R_3$$

$$\sim \begin{bmatrix} 1 & 0 & 0 & : & -\frac{1}{2} \\ 0 & 1 & 0 & : & -1 \\ 0 & 0 & 1 & : & 1 \end{bmatrix}$$

Hence, the matrix A is in normal form

$$\therefore x_1 = -\frac{1}{2}, x_2 = -1, x_3 = 1.$$

Example 15. Solve by Gauss-Jordan elimination method:

$$\begin{aligned} 2x_1 + 5x_2 + 2x_3 - 3x_4 &= 3 \\ 3x_1 + 6x_2 + 5x_3 + 2x_4 &= 2 \\ 4x_1 + 5x_2 + 14x_3 + 14x_4 &= 11 \\ 5x_1 + 10x_2 + 8x_3 + 4x_4 &= 4 \end{aligned}$$

Sol. $[A : B] = \begin{bmatrix} 2 & 5 & 2 & -3 & : & 3 \\ 3 & 6 & 5 & 2 & : & 2 \\ 4 & 5 & 14 & 14 & : & 11 \\ 5 & 10 & 8 & 4 & : & 4 \end{bmatrix}$

$$R_2 \rightarrow R_2 - R_1, R_3 \rightarrow R_3 - R_2, R_4 \rightarrow R_4 - R_3$$

$$\sim \begin{bmatrix} 2 & 5 & 2 & -3 & : & 3 \\ 1 & 1 & 3 & 5 & : & -1 \\ 1 & -1 & 9 & 12 & : & 9 \\ 1 & 5 & -6 & -10 & : & -7 \end{bmatrix}$$

$$R_1 \leftrightarrow R_4, R_2 \leftrightarrow R_3$$

$$\sim \begin{bmatrix} 1 & 5 & -6 & -10 & : & -7 \\ 1 & -1 & 9 & 12 & : & 9 \\ 1 & 1 & 3 & 5 & : & -1 \\ 2 & 5 & 2 & -3 & : & 3 \end{bmatrix}$$

$$R_2 \rightarrow R_2 - R_1, R_3 \rightarrow R_3 - R_1, R_4 \rightarrow R_4 - 2R_1$$

$$\sim \begin{bmatrix} 1 & 5 & -6 & -10 & : & -7 \\ 0 & -6 & 15 & 22 & : & 16 \\ 0 & -4 & 9 & 15 & : & 6 \\ 0 & -5 & 14 & 17 & : & 17 \end{bmatrix}$$

$R_2 \rightarrow R_2 - R_4$, then $R_2 \rightarrow -R_2$, $R_3 \rightarrow R_3 - 4R_2$, then again $R_3 \rightarrow -R_3$, $R_4 \rightarrow R_4 - 5R_2$, then also again $R_4 \rightarrow -R_4$

$$\sim \begin{bmatrix} 1 & 5 & -6 & -10 & : & -7 \\ 0 & 1 & -1 & -5 & : & 1 \\ 0 & 0 & -5 & 5 & : & -10 \\ 0 & 0 & 9 & -8 & : & 22 \end{bmatrix}$$

$$R_3 \rightarrow -\frac{1}{5}R_3, \text{ then } R_4 \rightarrow R_4 - 9R_3$$

$$\sim \begin{bmatrix} 1 & 5 & -6 & -10 & : & -7 \\ 0 & 1 & -1 & -5 & : & 1 \\ 0 & 0 & 1 & -1 & : & 2 \\ 0 & 0 & 0 & 1 & : & 4 \end{bmatrix}$$

$$R_3 \rightarrow R_3 + R_4, R_2 \rightarrow R_2 + 5R_4, R_1 \rightarrow R_1 + 10R_4$$

$$\sim \begin{bmatrix} 1 & 5 & -6 & 0 & : & 33 \\ 0 & 1 & -1 & 0 & : & 21 \\ 0 & 0 & 1 & 0 & : & 6 \\ 0 & 0 & 0 & 1 & : & 4 \end{bmatrix}$$

$$R_2 \rightarrow R_2 + R_3, R_1 \rightarrow R_1 + 6R_3$$

$$\sim \begin{bmatrix} 1 & 5 & 0 & 0 & : & 69 \\ 0 & 1 & 0 & 0 & : & 27 \\ 0 & 0 & 1 & 0 & : & 26 \\ 0 & 0 & 0 & 1 & : & 4 \end{bmatrix}$$

$$R_1 \rightarrow R_1 - 5R_2$$

$$\sim \begin{bmatrix} 1 & 0 & 0 & 0 & : & -66 \\ 0 & 1 & 0 & 0 & : & 27 \\ 0 & 0 & 1 & 0 & : & 6 \\ 0 & 0 & 0 & 1 & : & 4 \end{bmatrix}$$

Hence, the coefficient matrix is in normal form

$$\therefore x_1 = -66, x_2 = 27, x_3 = 6 \text{ and } x_4 = 4.$$

Example 16. Find the values of λ for which the equations

$$3x + y - \lambda z = 0; 4x - 2y - 3z = 0; 2\lambda x + 4y + \lambda z = 0$$

have a non-trivial solution. Obtain the most general solutions in each case.

Sol. The coefficient matrix is

$$A = \begin{bmatrix} 3 & 1 & -\lambda \\ 4 & -2 & -3 \\ 2\lambda & 4 & \lambda \end{bmatrix}$$

for non-trivial solution $|A| = 0$

$$\therefore |A| = \begin{vmatrix} 3 & 1 & -\lambda \\ 4 & -2 & -3 \\ 2\lambda & 4 & \lambda \end{vmatrix} = 0 \Rightarrow 3(-2\lambda + 12) - (4\lambda + 6\lambda) - \lambda(16 + 4\lambda) = 0$$

$$\Rightarrow -6\lambda + 36 - 10\lambda - 16\lambda - 4\lambda^2 = 0$$

$$\Rightarrow -4\lambda^2 - 32\lambda + 36 = 0 \Rightarrow \lambda^2 + 8\lambda - 9 = 0$$

$$\Rightarrow \lambda^2 + 9\lambda - \lambda - 9 = 0 \Rightarrow (\lambda - 1)(\lambda + 9) = 0$$

$$\Rightarrow \lambda = 1, -9.$$

Case I. If $\lambda = 1$,

$$A = \begin{bmatrix} 3 & 1 & -1 \\ 4 & -2 & -3 \\ 2 & 4 & 1 \end{bmatrix}$$

$$R_2 \rightarrow R_2 - \frac{4}{3}R_1, R_3 \rightarrow R_3 - \frac{2}{3}R_1$$

$$\sim \begin{bmatrix} 3 & 1 & -1 \\ 0 & -\frac{10}{3} & -\frac{5}{3} \\ 0 & \frac{10}{3} & \frac{5}{3} \end{bmatrix}$$

$$R_3 \rightarrow R_3 + R_2$$

$$\sim \begin{bmatrix} 3 & 1 & -1 \\ 0 & -\frac{10}{3} & -\frac{5}{3} \\ 0 & 0 & 0 \end{bmatrix}$$

This shows $\rho(A) = 2 < 3$ ($n = 3$)

$\therefore$ Let $z = k$

$$-\frac{10y}{3} - \frac{5z}{3} = 0 \Rightarrow 2y + k = 0 \Rightarrow y = -\frac{k}{2}$$

and $3x + y - z = 0 \Rightarrow 3x - \frac{k}{2} - k = 0 \Rightarrow x = \frac{k}{2}$. **Ans.**

Case II. If $\lambda = -9$

$$A = \begin{bmatrix} 3 & 1 & 9 \\ 4 & -2 & -3 \\ -18 & 4 & -9 \end{bmatrix}$$

Solve itself like case I.

$$x = -\frac{3k}{2}, y = -\frac{9k}{2}, z = k.$$

EXERCISE 3.4

Examine whether the following systems of equations are consistent. If consistent solve.

1.
$$\begin{aligned} x + y + z &= 6 \\ 2x + 3y - 2z &= 2 \\ 5x + y + 2z &= 13 \end{aligned} \quad (U.P.T.U., 2000) \text{ [Ans. } x = 1, y = 2, z = 3]$$
2.
$$\begin{aligned} 3x + 3y + 2z &= 1 \\ x + 2y &= 4 \end{aligned}$$

$$\begin{aligned} 10y + 3z &= -2 \\ 2x - 3y - z &= 5 \end{aligned} \quad [\text{Ans. } x = 2, y = 1 \text{ and } z = -4]$$

$$\begin{aligned} 3. \quad & x_1 + 2x_2 - x_3 = 3 \\ & 3x_1 - x_2 + 2x_3 = 1 \\ & 2x_1 - 2x_2 + 3x_3 = 2 \\ & x_1 - x_2 + x_3 = -1 \end{aligned} \quad (U.P.T.U., 2002) [\text{Ans. } x_1 = -1, x_2 = 4, x_3 = 4]$$

$$\begin{aligned} 4. \quad & x + y + z = 7 \\ & x + 2y + 3z = 8 \\ & y + 2z = 6 \end{aligned} \quad [\text{Ans. inconsistent; no solution}]$$

$$\begin{aligned} 5. \quad & 5x + 3y + 7z = 4 \\ & 3x + 26y + 2z = 9 \\ & 7x + 2y + 10z = 5 \end{aligned} \quad [\text{Ans. } x = (7 - 6k)/11, y = (3 + k)/11, z = k]$$

$$\begin{aligned} 6. \quad & -x_1 + x_2 + 2x_3 = 2 \\ & 3x_1 - x_2 + x_3 = 6 \\ & -x_1 + 3x_2 + 4x_3 = 4 \end{aligned} \quad [\text{Ans. } x_1 = 1, x_2 = -1, x_3 = 2]$$

7. Find the values of a and b for which the system has (i) no solution, (ii) unique solution and (iii) infinitely many solution.

$$\begin{aligned} 2x + 3y + 5z &= 9 \\ 7x + 3y - 2z &= 8 \\ 2x + 3y + az &= b \end{aligned} \quad \left[\begin{array}{ll} (i) & a = 5, b \neq 9 \\ \text{Ans. (ii)} & a \neq 5, b \text{ any value} \\ (iii) & a = 5, b = 9 \end{array} \right]$$

8. Discuss the solutions of the system of equations for all values of λ .

$$x + y + z = 2, 2x + y - 2z = 2, \lambda x + y + 4z = 2.$$

[Ans. Unique solution if $\lambda \neq 0$; infinite number of solutions if $\lambda = 0$]

$$\begin{aligned} 9. \quad & x + y + z = 6 \\ & x + 2y + 3z = 14 \\ & x + 4y + 7z = 30 \end{aligned} \quad [\text{Ans. } x = k - 2, y = 8 - 2k, z = k]$$

$$\begin{aligned} 10. \quad & 2x - y + z = 7 \\ & 3x + y - 5z = 13 \\ & x + y + z = 5 \end{aligned} \quad [\text{Ans. } x = 4, y = 1, z = 0]$$

Solve the following homogeneous equations:

$$\begin{aligned} 11. \quad & x + 2y + 3z = 0 \\ & 2x + y + 3z = 0 \\ & 3x + 2y + z = 0 \end{aligned} \quad [\text{Ans. } x = y = z = 0]$$

$$\begin{aligned} 12. \quad & x + y + 3z = 0 \\ & x - y + z = 0 \\ & x - 2y = 0 \\ & x - y + z = 0 \end{aligned} \quad [\text{Ans. } x = -2k, y = -k, z = k]$$

$$\begin{aligned} 13. \quad & x + 2y + 3z = 0 \\ & 3x + 4y + 4z = 0 \\ & 7x + 10y + 12z = 0 \end{aligned} \quad [\text{Ans. } x = y = z = 0]$$

14. $4x + 2y + z + 3w = 0$
 $6x + 3y + 4z + 7w = 0$
 $2x + y + w = 0$ [Ans. $x = k_1, y = -2k_1 - k_2, z = -k_2, w = k_2$]

15. For what values of λ the given equations will have a non-trivial solution.

$$\begin{aligned} x + 2y + 3z &= \lambda x \\ 2x + 3y + z &= \lambda x \\ 3x + y + 2z &= \lambda y \end{aligned} \quad [\text{Ans. } \lambda = 6]$$

16. Find the value of 'a' so that the following system of homogeneous equations have exactly 2 linearly independent solutions.

$$\begin{aligned} ax_1 - x_2 - x_3 &= 0 \\ -x_1 + ax_2 - x_3 &= 0 \\ -x_1 - x_2 + ax_3 &= 0 \end{aligned} \quad [\text{Ans. } a = -1]$$

17. Apply the test of rank to examine if the following equations are consistent:

$$\begin{aligned} 2x - y + 3z &= 8 \\ -x + 2y + z &= 4 \\ 3x + y - 4z &= 0 \end{aligned}$$

and if consistent, find the complete solution.

[Ans. $x = y = z = 2$]

18. Show that the equations

$$\begin{aligned} -2x + y + z &= a \\ x - 2y + z &= b \\ x + y - 2z &= c \end{aligned}$$

have no solutions unless $a + b + c = 0$, in which case they have infinitely many solutions.

Find their solutions $a = 1, b = 1$ and $c = -2$. [Ans. $x = k - 1, y = k - 1, z = k$]

19. Show that the system of equations $x + 2y - 2z = 0, 2x - y - w = 0, x + 2y - w = 0, 4x - y + 3z - w = 0$ do not have a non-trivial solution.

20. Show that the homogeneous system of equations $x + y \cos \gamma + z \cos \beta = 0, x \cos \gamma + y + z \cos \alpha = 0, x \cos \beta + y \cos \gamma + z = 0$, has non-trivial solution if $\alpha + \beta + \gamma = 0$.

Solve the following system of equations by Gaussian elimination method:

21. $x_1 + 2x_2 - x_3 = 3$
 $2x_1 - 2x_2 + 3x_3 = 2$
 $3x_1 - x_2 + 2x_3 = 1$
 $x_1 - x_2 + x_3 = -1$ [Ans. $x_1 = 1, x_2 = 4, x_3 = 4$]

22. $2x_1 + x_2 + x_3 = 10$
 $3x_1 + 2x_2 + 3x_3 = 18$
 $x_1 + 4x_2 + 9x_3 = 16$ [Ans. $x_1 = 7, x_2 = -9, x_3 = 5$]

23. $2x_1 + x_2 + 4x_3 = 12$
 $8x_1 - 3x_2 + 2x_3 = 20$
 $4x_1 + 11x_2 - x_3 = 33$ [Ans. $x_1 = 3, x_2 = 2, x_3 = 1$]

$$\begin{aligned} 24. \quad & x_1 + 4x_2 - x_3 = -5 \\ & x_1 + x_2 - 6x_3 = -12 \\ & 3x_1 - x_2 - x_3 = 4 \end{aligned}$$

$$\left[\text{Ans. } x_1 = \frac{117}{71}, x_2 = -\frac{81}{71}, x_3 = \frac{148}{71} \right]$$

$$\begin{aligned} 25. \quad & 2x_1 + x_2 + 2x_3 + x_4 = 6 \\ & 6x_1 - 6x_2 + 6x_3 + 12x_4 = 36 \\ & 4x_1 + 3x_2 + 3x_3 - 3x_4 = -1 \end{aligned}$$

$$2x_1 + 2x_2 - x_3 + x_4 = 10$$

$$[\text{Ans. } x_1 = 2, x_2 = 1, x_3 = -1, x_4 = 3]$$

$$\begin{aligned} 26. \quad & 2x_1 - 7x_2 + 4x_3 = 9 \\ & x_1 + 9x_2 - 6x_3 = 1 \\ & -3x_1 + 8x_2 + 5x_3 = 6 \end{aligned}$$

$$[\text{Ans. } x_1 = 4, x_2 = 1, x_3 = 2]$$

Solve by Gauss-Jordan method:

$$\begin{aligned} 27. \quad & x - 3y - 8z = -10 \\ & 3x + y = 4 \end{aligned}$$

$$2x + 5y + 6z = 13$$

$$[\text{Ans. } x = y = z = 1]$$

$$\begin{aligned} 28. \quad & x + y + z = 6 \\ & 2x + 3y - 2z = 2 \\ & 5x + y + 2z = 13 \end{aligned}$$

$$[\text{Ans. } x = 1, y = 2, z = 3]$$

$$\begin{aligned} 29. \quad & 3x + y + 2z = 3 \\ & 2x - 3y - z = -3 \\ & x + 2y + z = 4 \end{aligned}$$

$$[\text{Ans. } x = 1, y = 2, z = -1]$$

$$\begin{aligned} 30. \quad & 4x + 3y + 3z = -2 \\ & x + z = 0 \\ & 4x + 4y + 3z = -3 \end{aligned}$$

$$[\text{Ans. } x = 1, y = 1, z = -1]$$

$$31. \text{ Test the consistency and solve } 2x - 3y + 7z = 5, 3x + y - 3z = 13, 2x + 19y - 47z = 32.$$

$$[\text{Ans. Inconsistent}]$$

Solve the following system by any method:

$$\begin{aligned} 32. \quad & 2x + 6y + 7z = 0 \\ & 6x + 20y - 6z = -3 \\ & 6y - 18z = -1 \end{aligned}$$

$$[\text{Ans. Inconsistent}]$$

$$\begin{aligned} 33. \quad & 4x - y + 6z = 16 \\ & x - 4y - 3z = -16 \\ & 2x + 7y + 12z = 48 \\ & 5x - 5y + 3z = 0 \end{aligned}$$

$$\left[\text{Ans. } x = \frac{-9k}{5} + \frac{16}{3}, y = \frac{-6k}{5} + \frac{16}{3}, z = k \right]$$

$$\begin{aligned} 34. \quad & 2x_1 + x_2 + 5x_3 + x_4 = 5 \\ & x_1 + x_2 - 3x_3 - 4x_4 = -1 \\ & 3x_1 + 6x_2 - 2x_3 + x_4 = 8 \\ & 2x_1 + 2x_2 + 2x_3 - 3x_4 = 2 \end{aligned}$$

$$\left[\text{Ans. } x_1 = 2, x_2 = \frac{1}{5}, x_3 = 0, x_4 = \frac{4}{5} \right]$$

$$\begin{aligned} 35. \quad & 5x + 3y + 7z = 4 \\ & 3x + 26y + 2z = 9 \\ & 7x + 2y + 11z = 5 \end{aligned}$$

$$\left[\text{Ans. } x = \frac{7}{11}, y = \frac{3}{11}, z = 0 \right]$$